Chapter 10

Periodic Orbits

A solution $\gamma\colon \mathbb{R} \to \mathbb{R}^L$ to a differential equation $\dot{x} = f(t,x)$, $x \in \mathbb{R}^L$, is *periodic* if there exists $\tau > 0$ such that $\gamma(\tau) = \gamma(0)$. The goal of this chapter is to demonstrate how the Radii Polynomial Method, Theorem 7.6.2, can be used to prove the existence of periodic solutions to smooth systems of ODEs. We begin in Section 10.1 by considering a periodically forced second order scalar ODE as this allows us to present the general functional analytic framework without worrying about the period τ or the notational technicalities associated with systems of ODEs. We consider the general case of periodic orbits in polynomial vector fields in Section 10.2. As in the case of fixed points, periodic orbits often arise in systems via bifurcations, the most studied of which is the Hopf bifurcation. In Section 10.4 we indicate how we can rigorously identify Hopf bifurcations.

10.1 Example: a scalar equation with forcing

Consider the Helmholtz-Duffing type equation

$$u''(t) + c_1 u'(t) + c_2 u(t) - u(t)^2 = c_3 \cos t, \tag{10.1}$$

where c_1, c_2, c_3 are parameters. With $c_1 > 0$ and $c_3 = 0$, equation (10.1) is a model for a damped nonlinear spring whose phase portrait is shown in Figure **??**. Observe that there are two equilibria $u = 0$ and $u = 1/c_2$. The magnitude of the parameter c_3 indicates the intensity with which the system is periodically forced. For c_3 sufficiently small, perturbation theory can be use to show that the periodic forcing transforms the fixed points into periodic orbits. Our goal is to demonstrate the existence of the periodic orbits for fixed values of $|c_3|$.

Our strategy of proof is to use Theorem 7.6.2 the first step of which is to identify the appropriate Banach spaces. Sine and cosine provide a natural choice of functions by which a periodic orbit can be identified. With this in mind we turn our attention to the space of periodic functions expressed in terms of Fourier series thus we will consider functions of the form

$$u(t) = \sum_{n\in\mathbb{Z}} a_n e^{int} \tag{10.2}$$

where $a = \{a_n \in \mathbb{C}\}_{n\in\mathbb{Z}} \in \ell^1_{\nu,\mathbb{Z}}$. Observe that even though we are interested in periodic solutions that take real values we work in the space of complex valued function. Furthermore, note that we have chosen the period of u to coincide with the period of $\cos t$ in (10.1). For autonomous problems the period of the periodic orbit is an unknown and therefore needs to be solved for.

Set $X = \ell^1_{\nu,\mathbb{Z}}$ where we emphasize that the sequences take complex values. To identify the desired zero finding problem we substitute (10.2) into (10.1), observe that $\cos t = (e^{it} + e^{-it})/2$, and match powers to obtain

$$F_n(a) \stackrel{\text{def}}{=} [-n^2 + inc_1 + c_2]a_n - (a * a)_n - c_3 b_n,$$

where

$$b_n = \begin{cases} \frac{1}{2} & \text{if } n = \pm 1, \\ 0 & \text{otherwise.} \end{cases}$$

To identify an appropriate Banach space for the image of F we make use of the following proposition, the proof of which involves a direct computation.

Proposition 10.1.1. *For fixed $\nu > 0$ let $\omega = \left\{ \frac{\nu^{|n|+2}}{|n|(|n|+1)} \right\}_{n\in\mathbb{Z}}$ be a sequence of weights. If $a = \{a_n\}_{n\in\mathbb{Z}} \in \ell^1_\nu$, then $a \in \ell^1_\omega$ and $\left\{(-n^2 + inc_1)a_n\right\}_{n\in\mathbb{Z}} \in \ell^1_\omega$.*

Let ω be the sequence of weights in Proposition 10.1.1 and set $Y = \ell^1_{\omega,\mathbb{Z}}$. It is left to the reader to check that $F\colon X \to Y$.

The second step in application of Theorem 7.6.2 is the identification of the approximate solution $\bar{a} \in X$. As is done in the previous sections this is obtained by truncating F. However, we are using the Cauchy product on $\ell^1_{\nu,\mathbb{Z}}$ and thus the formula for the truncation is slightly different. Define $F^{(N)}$ by

$$F^{(N)}_n(a) \stackrel{\text{def}}{=} [-n^2 + inc_1 + c_2]a_n - \sum_{\substack{j_1+j_2=n \\ |j_1|,|j_2|\le N}} a_{j_1}a_{j_2} - c_3 b_n. \tag{10.3}$$

Let $\bar{a}^{(N)} = (\bar{a}^{(N)}_{-N}, \bar{a}^{(N)}_{-N+1}, \ldots, \bar{a}^{(N)}_{N-1}, \bar{a}^{(N)}_{N})$ be an approximate zero of $F^{(N)}$ and observe that

$$\bar{u}^{(N)}(t) \stackrel{\text{def}}{=} \sum_{k=-N}^{N} \bar{a}_k e^{ikt}$$

is a periodic function. However, to be an approximate solution to (10.1) we desire that $\bar{u}^{(N)}$ be a real valued function, and therefore, we require

$$\bar{a}_{-n} = \bar{a}_n^* \quad \text{for all } n = -N, \dots, N$$

where * denotes complex conjugation. Define

$$\bar{a} \stackrel{\text{def}}{=} (\dots, 0, 0, 0, \bar{a}_{-N}^{(N)}, \bar{a}_{-N+1}^{(N)}, \dots \bar{a}_{N-1}^{(N)}, \bar{a}_N^{(N)}, 0, 0, 0, \dots) \in X. \tag{10.4}$$

The third step in application of Theorem 7.6.2 is the choice of $A^\dagger \in B(X, Y)$ that approximates $DF(\bar{a})$. Observe that

$$(DF(a))_{n,j} = \frac{\partial F_n}{\partial a_j}(a) = [-n^2 + inc_1 + c_2]\delta_{n,j} - 2a_{n-j}.$$

If N is large and $|j|$ is small compared to $|n| > N$, then $|a_{n-j}|$ is small, suggesting that $A^\dagger$ be defined by ignoring the off diagonal terms for $|n| > N$. Visualizing $A^\dagger$ as an infinite matrix leads to

$$A^\dagger = \begin{bmatrix} \Lambda_- & 0 & 0 \\ 0 & DF^{(N)}(\bar{a}^{(N)}) & 0 \\ 0 & 0 & \Lambda_+ \end{bmatrix}$$

where Λ_- and Λ_+ are diagonal matrices with diagonal entries $-n^2 + inc_1 + c_2$, where $n < -N$ and $n > N$, respectively. To be more explicit, the action of $A^\dagger$ on a vector $h \in X$ is given by

$$(A^\dagger h)_n \stackrel{\text{def}}{=} \begin{cases} [DF^{(N)}(\bar{a}^{(N)})h^{(N)}]_n, & |n| \le N \\ [-n^2 + inc_1 + c_2]h_n, & |n| > N. \end{cases} \tag{10.5}$$

Here $h^{(N)} = (h_{-N}, \dots, h_N) \in \mathbb{C}^{2N+1}$ is obtained by projecting h onto its center $(2N+1)$ coordinates.

It is left to the reader to check that $A^\dagger \in B(X, Y)$.

The fourth step is to choose $A \in B(Y, X)$ an approximate inverse of $A^\dagger$. Let $A^{(N)}$ be an approximate inverse of $DF^{(N)}(\bar{a}^{(N)})$. Then, A is obtained using $A^{(N)}$ and $\Lambda_\pm^{-1}$, which using the matrix representation has the form

$$A \begin{bmatrix} \Lambda_-^{-1} & 0 & 0 \\ 0 & A^{(N)} & 0 \\ 0 & 0 & \Lambda_+^{-1} \end{bmatrix}.$$

It is left to the reader to check that $A \in B(Y, X)$.

To apply Theorem 7.6.2 we need the Y_0, Z_0, Z_1, and Z_2 bounds, which are collected in Theorem 10.1.3. It is important to note, however, that at this point a successful application of Theorem 7.6.2 proves the existence a unique periodic function $\tilde{u} \colon \mathbb{R} \to \mathbb{C}$ that satisfies (10.1). We need to show that $\tilde{u} \colon \mathbb{R} \to \mathbb{R}$. Since $\tilde{u}(t) \stackrel{\text{def}}{=} \sum_{n \in \mathbb{Z}} \tilde{a}_n e^{int}$ it is sufficient to prove that $\tilde{a}_{-n} = \tilde{a}_n^*$.

Proposition 10.1.2. *Assume* $\tilde{a} = \{\tilde{a}_n\}_{n\in\mathbb{Z}} \in X$ *is the unique fixed point in* $B_{r_0}(\bar{a})$ *of the map* $T(a) = a - AF(a)$. *Then,* $\tilde{a}_{-n} = \tilde{a}_n^*$ *for all* $n \in \mathbb{Z}$.

Proof. Set $\hat{a}_n = \tilde{a}_{-n}^*$. By assumption $\tilde{a} \in X$, and therefore $\hat{a} \in X$. Recall that $\bar{a}_n = \bar{a}_{-n}^*$, for all $|n| \leq N$. Therefore,

$$\begin{aligned}\|\hat{a} - \bar{a}\|_{1,\nu} &= \sum_{n\in\mathbb{Z}} |\hat{a}_n - \bar{a}_n|\nu^{|n|} = \sum_{n\in\mathbb{Z}} |\tilde{a}_{-n}^* - \bar{a}_{-n}^*|\nu^{|n|} \\ &= \sum_{n'\in\mathbb{Z}} |\tilde{a}_{n'}^* - \bar{a}_n^*|\nu^{|n'|} = \|\tilde{a} - \bar{a}\|_{1,\nu}.\end{aligned}$$

This implies that $\hat{a} \in B_{r_0}(\bar{a})$. Moreover, a short calculation shows that

$$F(\hat{a})_{-n}^* = F(\tilde{a})_n,$$

where one uses that $b_{-n} = b_n^*$ and

$$(\hat{a} * \hat{a})_{-n}^* = \sum_{j\in\mathbb{Z}} \hat{a}_{-n-j}^* \hat{a}_j^* = \sum_{j\in\mathbb{Z}} \tilde{a}_{n+j}^* \tilde{a}_{-j} = \sum_{j'\in\mathbb{Z}} \tilde{a}_{n-j'}^* \tilde{a}_{j'} = (\tilde{a} * \tilde{a})_n.$$

Since $\tilde{a}$ is a zero of F, it follows that $F(\hat{a}) = 0$, and hence $T(\hat{a}) = \hat{a}$. By uniqueness of the fixed point of T in $B_{r_0}(\bar{a})$ we conclude that $\hat{a} = \tilde{a}$. This is equivalent with $\tilde{a}_{-n} = \tilde{a}_n^*$ for all $n \in \mathbb{Z}$. □

Theorem 10.1.3. *Fix* $c_1, c_2, c_3 \in \mathbb{R}$, $\nu > 1$ *and* $N \in \mathbb{N}$ *such that* $(N+1)^2 > c_2$. *Let* $F^{(N)} \colon \mathbb{C}^{2N+1} \to \mathbb{C}^{2N+1}$ *be defined by* (10.3). *Let* $\bar{a}^{(N)} = (\bar{a}_{-N}, \ldots, \bar{a}_N)$ *with* $\bar{a}_{-n} = \bar{a}_n^*$ *be an approximate solution to* $F^{(N)}(a) = 0$ *and let* $\bar{a}$ *be given by* (10.4). *Let* $A^{(N)}$ *be a numerical inverse of* $DF^{(N)}(\bar{a}^{(N)})$. *Set*

$$\begin{aligned}
Y_0 &\stackrel{\text{def}}{=} \sum_{n=-N}^{N} \left| \left(A^{(N)} F^{(N)}(\bar{a}) \right)_n \right| \nu^{|n|} + \sum_{|n|=N+1}^{2N} \frac{\left|(\bar{a} * \bar{a})_n\right| \nu^{|n|}}{|n^2 - inc_1 - c_2|}, \\
Z_0 &\stackrel{\text{def}}{=} \|I - A^{(N)} DF^{(N)}(\bar{a}^{(N)})\|_{1,\nu,N}, \\
Z_1 &\stackrel{\text{def}}{=} 2 \left[\|A^{(N)} \Gamma_\nu^{(N)}\|_{1,\nu,N} + \frac{1}{|(N+1)^2 - i(N+1)c_1 - c_2|} \right] \|\bar{a}\|_{1,\nu}, \\
Z_2 &\stackrel{\text{def}}{=} 2 \max \left\{ \|A^{(N)}\|_{1,\nu,N}, \frac{1}{|(N+1)^2 - i(N+1)c_1 - c_2|} \right\}
\end{aligned}$$

where $\Gamma_\nu^{(N)} \in M_{2N+1}(\mathbb{C})$ *is the diagonal matrix with diagonal elements*

$$(\Gamma_\nu^{(N)})_{n,n} = \nu^{2|n| - 2(N+1)}. \tag{10.6}$$

Then, Y_0, Z_0, Z_1, *and* Z_2 *satisfy the hypotheses of Theorem 7.6.2.*

We leave the estimates Y_0, Z_0 and Z_2 to the reader as these are completely analogous to those in Section 8.1. The Z_1 estimate in the case of Fourier series is considerably more involved than in the case of Taylor series, hence we provide the details below.

As a first step we provide a useful partial bound for the Cauchy product in the setting of $\ell^1_{\nu,\mathbb{Z}}$.

Proposition 10.1.4. *Let $a, h \in \ell^1_{\nu,\mathbb{Z}}$. Let $N \in \mathbb{N}$ and assume $h_n = 0$ for $|n| \leq N$. Then*

$$\sum_{|n|\leq N} |(a*h)_n|\nu^{-|n|} \leq \nu^{-2N-2}\|a\|_{1,\nu}\|h\|_{1,\nu}.$$

Proof. It follows from an application of the triangle inequality that

$$\begin{aligned}
\sum_{|n|\leq N} |(a*h)_n|\nu^{-|n|} &= \sum_{|n|\leq N} \left| \sum_{j\in\mathbb{Z}} a_{n-j} * h_j \right| \nu^{-|n|} \\
&\leq \sum_{|n|\leq N} \sum_{|j|>N} |a_{n-j}||h_j||\nu^{-|n|} \\
&\leq \sum_{|j|>N} |h_j|\nu^{|j|} \sum_{|n|\leq N} |a_{n-j}|\nu^{-|n|-|j|} \\
&\leq \|h\|_{1,\nu} \sup_{|j|>N} \sum_{|n|\leq N} |a_{n-j}|\nu^{-|n|-|j|}.
\end{aligned}$$

For any $|j| > N$,

$$\begin{aligned}
\sum_{|n|\leq N} |a_{n-j}|\nu^{-|n|-|j|} &= \sum_{|n|\leq N} |a_{n-j}|\nu^{|-j|}\nu^{-|n|-|j|-|n-j|} \\
&\leq \|\phi\|_{1,\nu} \sup_{|j|>N, |n|\leq N} \nu^{-|n|-|j|-|n-j|}.
\end{aligned} \tag{10.7}$$

Since for all $|j| > N$ and $|n| \leq N$ we have

$$|n| + |j| + |n-j| = |n| + |j| + |j| - |n| = 2|j| \geq 2(N+1),$$

we arrive at the asserted estimate. $\square$

We return to the proof of the Z_1 estimate. We need to show that

$$\|A(DF(\bar{a}) - A^\dagger)\|_{B(\ell^1_\nu)} \leq Z_1, \tag{10.8}$$

where Z_1 is given in Theorem 10.1.3. Given $h \in X$ with $\|h\|_{1,\nu} \leq 1$ we set

$$w \stackrel{\text{def}}{=} [A(DF(\bar{a}) - A^\dagger)]h$$

and estimate

$$w^{(N)} = (w_{-N}, \ldots, w_N) \in \mathbb{C}^{2N+1}$$

and

$$w^{\infty} = (\ldots, w_{-N-2}, w_{-N-1}, 0, 0, \ldots, 0, 0, w_{N+1}, w_{N+2}, \ldots) \in \ell^1_\nu,$$

separately.

Lemma 10.1.5. *For any $h \in X$ with $\|h\|_{1,\nu} \leq 1$ and w, $w^{(N)}$ and w^{∞} given as above we have*

$$\begin{aligned}
\|w^{(N)}\|_{1,\nu} &\leq 2\|A^{(N)}\Gamma_\nu^{(N)}\|_{1,\nu,N}\|\bar{a}\|_{1,\nu} \\
\|w^{\infty}\|_{1,\nu} &\leq \frac{2}{|(N+1)^2 - i(N+1)c_1 - c_2|}\|\bar{a}\|_{1,\nu}.
\end{aligned}$$

Since $\|w\|_{1,\nu} = \|w^{(N)}\|_{1,\nu} + \|w^{\infty}\|_{1,\nu}$ the estimate (10.8) holds for Z_1 given in Theorem 10.1.3.

Proof. Define $z^{(N)} \in \mathbb{C}^{2N+1}$ by

$$z_n^{(N)} \stackrel{\text{def}}{=} ((DF(\bar{a}) - A^{\dagger})h)_n \quad \text{for } |n| \leq N.$$

Note that it follows from (10.5), the definition of $A^{\dagger}$, that

$$z_n^{(N)} = \sum_{j\in\mathbb{Z}} 2\bar{a}_{n-j}h_j - \sum_{|j|\leq N} 2\bar{a}_{n-j}h_j = \sum_{|j|>N} 2\bar{a}_{n-j}h_j = 2(\bar{a} * h^{\infty})_k,$$

where

$$h^{\infty} \stackrel{\text{def}}{=} (\ldots, h_{-N-2}, h_{-N-1}, 0, 0, \ldots, 0, 0, h_{N+1}, h_{N+2}, \ldots).$$

Observe that

$$w^{(N)} = A^{(N)}[(DF(\bar{a}) - A^{\dagger})h]^{(N)} = A^{(N)}z^{(N)} = A^{(N)}\Gamma_\nu^{(N)}(\Gamma_\nu^{(N)})^{-1}z^{(N)}.$$

It follows from the definition of $\Gamma^{(N)}$, Proposition 10.1.4, and the assumption $\|h\|_{1,\nu} \leq 1$ that

$$\begin{aligned}
\|(\Gamma_\nu^{(N)})^{-1}z^{(N)}\|_{1,\nu} &= \sum_{|n|\leq N} \nu^{2N+2-2|n|}|z_n^{(N)}|\nu^{|n|} \\
&= \nu^{2N+2}\sum_{|n|\leq N} 2|(\bar{a} * h^{\infty})_n|\nu^{-|n|} \leq 2\|\bar{a}\|_{1,\nu}.
\end{aligned}$$

We infer that

$$\|w^{(N)}\|_{1,\nu} \leq \|A^{(N)}\Gamma_\nu^{(N)}\|_{1,\nu,N}\|(\Gamma_\nu^{(N)})^{-1}y\|_{1,\nu} \leq 2\|A^{(N)}\Gamma_\nu^{(N)}\|_{1,\nu,N}\|\bar{a}\|_{1,\nu}.$$

Turning to the bound for $\|w^\infty\|_{1,\nu}$ observe that

$$w_n^\infty = \frac{2(\bar{a} * h)_n}{-n^2 + inc_1 + c_2} \qquad \text{for all } |n| > N.$$

Hence, it follows that

$$\begin{aligned}
\|w^\infty\|_{1,\nu} &= \sum_{|n|>N} |z_n| \nu^{|n|} \\
&= \sum_{|n|>N} \frac{2|(\bar{a} * h)_n| \nu^{|n|}}{|-k^2 + ikc_1 + c_2|} \\
&\le \frac{2}{|(N+1)^2 - i(N+1)c_1 - c_2|} \sum_{|n|>N} |(\bar{a} * h)_n| \nu^{|n|} \\
&\le \frac{2}{|(N+1)^2 - i(N+1)c_1 - c_2|} \|\bar{a} * h\|_{1,\nu} \\
&\le \frac{2}{|(N+1)^2 - i(N+1)c_1 - c_2|} \|\bar{a}\|_{1,\nu} \|h\|_{1,\nu} \\
&\le \frac{2}{|(N+1)^2 - i(N+1)c_1 - c_2|} \|\bar{a}\|_{1,\nu}.
\end{aligned}$$

□

We now turn to an application of Theorem 10.1.3. We take $c_1 = \frac{1}{10}$, $c_2 = 4$ and $c_3 = 1$. We choose $N = 10$ and $\nu = 1.2$. Then we obtain

$$Y_0 = 2.23 \cdot 10^{-9}, \quad Z_0 = 4.05 \cdot 10^{-14}, \quad Z_1 = 0.588, \quad Z_2 = 12.5.$$

For these values $p(5.6 \cdot 10^{-9}) < 0$ and hence we set $r_0 = 5.6 \cdot 10^{-9}$. Therefore, by Theorem there exists a solution $\tilde{u}$ of (10.1) corresponding to the unique zero $\tilde{a} \in B_{r_0}(\bar{a})$ of F. We have the error bound

$$\left\| \tilde{u}(t) - \sum_{k=-10}^{10} \bar{a}_k e^{ikt} \right\| \le r_0.$$

The solution, or rather the approximation $\bar{x}(t)$, is depicted in Figure 10.1.

10.2 Periodic solutions of polynomial vector fields

The focus of this section is to demonstrate how the Radii Polynomial Approach can be used to prove the existence of periodic solutions to autonomous systems of ODEs

$$\dot{x} = f(x), \quad x \in \mathbb{R}^L,$$

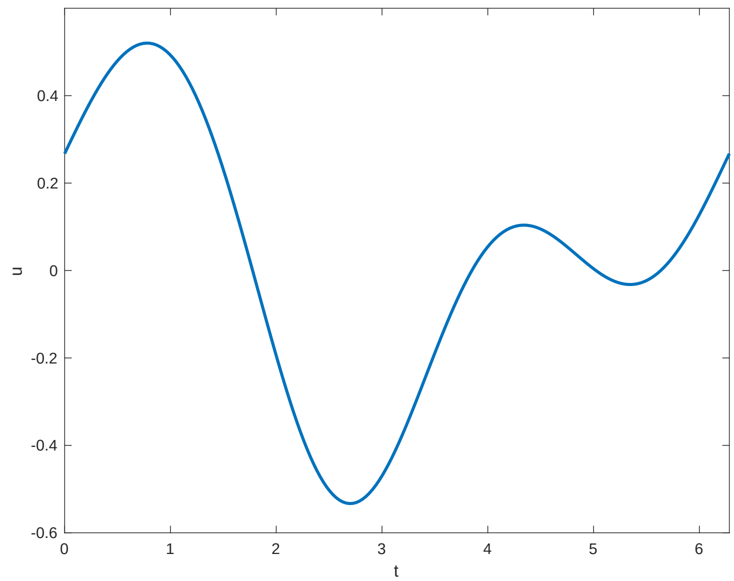

Figure 10.1: A periodic solution of (10.1) with $c_1 = \frac{1}{10}$, $c_2 = 4$ and $c_3 = 1$ lies within distance $r_0 = 5.6 \cdot 10^{-9}$ of the depicted graph.

where f is given in terms of polynomials. While the approach presented in this section and that of Section 10.1 are essentially the same there are some essential technical differences.

First, for autonomous systems we do not a priori know the period τ of the desired solution and we find working with τ-periodic functions where τ is part of the set of unknowns to be inconvenient. Hence we start by rescaling time by $\tau_0 = \frac{\tau}{2\pi}$, which transforms the problem into

$$\dot{x} = \tau_0 f(x) \tag{10.9}$$

and we restate our goal to be: prove the existence of a 2π-periodic function solving (10.9) for some (unknown) $\tau_0 > 0$.

Second, a fundamental conclusion of the Radii Polynomial Approach is the existence of a unique solution. There are two ways in which uniqueness can fail in the setting of autonomous systems and to address this it is helpful to think about periodic solutions geometrically. With this in mind assume that X is a topological space and that $\varphi\colon \mathbb{R}\times X \to X$ is a flow.

Definition 10.2.1. A point $x_0 \in X$ is a *periodic point* if there exists $\tau > 0$ such that $\varphi(\tau, x_0) = x_0$. The associate *periodic orbit* is $\Gamma \stackrel{\text{def}}{=} \varphi([0,\tau], x_0) \subset X$. The smallest $\tau > 0$ (if it exists) is called the *period* of the periodic orbit. The periodic orbit Γ is *isolated* if there exists a neighborhood $N \subset X$ of Γ such that Γ is the unique periodic orbit of φ in N.

The harmonic oscillator $\ddot{x} = -x$ provides a simple example of failure of isolation;

all solutions are periodic with period 2π, but none of the periodic orbits are isolated. The obvious conclusion is that we should not expect to be able to identify these periodic solutions using the Radii Polynomial Approach. However, isolation of periodic orbits in an of itself is not sufficient. To see this let $\gamma\colon \mathbb{R} \to \mathbb{R}^L$ be a 2π-periodic solution and let $\Gamma \stackrel{\text{def}}{=} \gamma(\mathbb{R})$ be an isolated periodic orbit. Define $\gamma_\theta(t) \stackrel{\text{def}}{=} \gamma(t+\theta)$ for all $\theta \in \mathbb{R}$. Observe that γ_θ is a 2π-periodic solution and $\Gamma = \gamma_\theta(\mathbb{R})$. In summary, a periodic orbit that is isolated in phase space does not give rise to an isolated periodic solution in function space.

One way to isolate the periodic solution is to set a *Poincaré phase condition.* More specifically, any application of Theorem 7.6.2 requires the existence of an approximate solution. Thus we can assume that we have an approximate 2π periodic orbit $\bar{\Gamma}$. Choose $\xi \in \bar{\Gamma}$. The idealized associated Poincaré phase condition is

$$f(\xi) \cdot (\gamma(0) - \xi) = 0, \tag{10.10}$$

where $\gamma\colon \mathbb{R} \to \mathbb{R}^L$ is the desired periodic solution (see Figure 10.2). We add the adjective idealized because to ensure uniqueness in function space it is not necessary that this exact condition be satisfied; we will exploit this freedom shortly.

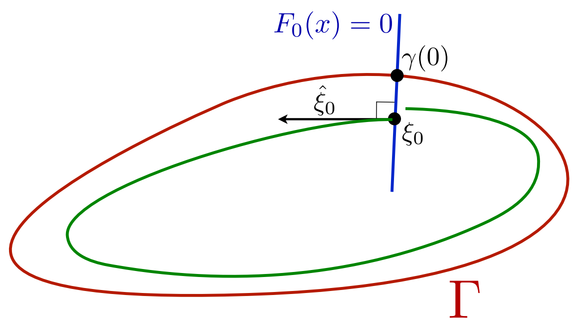

Figure 10.2: Poincaré phase condition. (need to modify)

We now turn to the technicalities of applying Theorem 7.6.2 to prove the existence of a 2π periodic solution to (10.9). As in Section 10.1 we express the desired solution $\gamma\colon \mathbb{R} \to \mathbb{R}^L$ using Fourier series

$$\gamma_l(t) = \sum_{n\in\mathbb{Z}} a_{l,n} e^{int}, \qquad l = 1, \ldots, L.$$

Since $f\colon \mathbb{R}^L \to \mathbb{R}^L$ is given in terms of polynomials there exist an integer $M \geq 0$ such that

$$f(x) = \sum_{|\alpha|=0}^{M} c_\alpha x^\alpha \quad \text{with } c_\alpha \in \mathbb{R}^L, \tag{10.11}$$

where $\alpha = (\alpha_1, \dots, \alpha_L)$ is a multi-index.

The l-th component of $f(\gamma(t))$ takes the form

$$f_l(\gamma(t)) = \sum_{n\in\mathbb{Z}} (\phi_l(a))_n e^{int}, \tag{10.12}$$

where the n-th Fourier coefficient $(\phi_l(a))_n$ is a combination of discrete convolutions involving the terms of a, i.e.,

$$\phi_l(a) = \sum_{|\alpha|=0}^{M} c_{l,\alpha}\, a_1^{\alpha_1} * \cdots * a_L^{\alpha_L}.$$

Set $\phi(a) = (\phi_1(a), \dots, \phi_L(a))$.

Using this notation we employ (10.9) to define the function $F\colon X \to Y$ by

$$(F_l(\tau_0, a))_n \stackrel{\text{def}}{=} ina_{l,n} - \tau_0(\phi_l(a))_n. \tag{10.13}$$

for all $n \in \mathbb{Z}$ and $l = 1, \dots, L$. To incorporate the idealized Poincaré phase condition we fix $\xi \in \mathbb{R}^L$ and set $\zeta = f(\xi)$. Expressing (10.10) using the Fourier series adds the constraint

$$\begin{aligned} \zeta \cdot (\gamma(0) - \xi) &= 0 \\ \sum_{l=1}^{L} (\zeta_l \gamma_l(0) - \zeta_l \xi_l) &= 0 \\ \sum_{l=1}^{L} \left(\zeta_l \sum_{n\in\mathbb{Z}} a_n - \zeta_l \xi_l \right) &= 0. \end{aligned} \tag{10.14}$$

For the purpose of computations we do not want to work with the infinite sum and instead we pick $N_0 \geq 0$ and use the constraint

$$F_0(\tau_0, a) \stackrel{\text{def}}{=} \sum_{l=1}^{L} \left(\zeta_l \sum_{|n|\leq N_0} a_n - \zeta_l \xi_l \right). \tag{10.15}$$

Observe that we have reduced the problem of proving the existence of periodic solutions of general polynomial vector fields to solving the operator equation

$$F(\tau_0, a) = \begin{pmatrix} F_0(\tau_0, a) \\ F_1(\tau_0, a) \\ \vdots \\ F_L(\tau_0, a) \end{pmatrix} = 0. \tag{10.16}$$

Writing F in this form makes it clear that $X = \mathbb{C} \times (\ell^1_{\nu,\mathbb{Z}})^L$.

We state the following proposition, whose proof is presented in Section 10.5, that guarantees that if we prove the existence of a zero of F, then we have proven the existence of a periodic solution to (10.9).

Proposition 10.2.2. *Let $\nu > 1$. Let $(\tau_0, a) \in \mathbb{C} \times (\ell^1_\nu)^L$ be such that $\tau_0 > 0$ and $a^*_{l,-n} = a_{l,n}$ for all $n \in \mathbb{Z}$ and $l = 1, \ldots, L$. If $F(\tau_0, a) = 0$, then*

$$\gamma(t) \stackrel{\text{def}}{=} \sum_{n \in \mathbb{Z}} a_n e^{int} \tag{10.17}$$

is a τ_0 periodic solution to (10.9).

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Lemma 10.2.3. *Let $\nu > 1$. Let $x = (\tau_0, a) \in \mathbb{C} \times (\ell^1_\nu)^n$ be such that $\tau_0 \in \mathbb{R}$ and $(a_j)^*_{-k} = (a_j)_k$ for all $k \in \mathbb{Z}$ and $j = 1, \ldots, n$. If x satisfies $F(x) = 0$, then*

$$\gamma(t) \stackrel{\text{def}}{=} \sum_{k \in \mathbb{Z}} a_k e^{ikt/\tau_0} \tag{10.18}$$

satisfies $\gamma'(t) = f(\gamma(t))$, i.e., it is a periodic orbit of the vector field f with period $2\pi\tau_0$.

Proof. When $F(x) = 0$, in particular we have

$$ik(a_j)_k = \tau_0(\phi_j(a))_k \qquad \text{for all } k \in \mathbb{Z} \text{ and } j = 1, \ldots, n. \tag{10.19}$$

For each $j = 1, \ldots, n$, since $a_j \in \ell^1_\nu$ and $\nu > 1$ the Fourier series

$$\hat{\gamma}_j(t) \stackrel{\text{def}}{=} \sum_{k \in \mathbb{Z}} (a_j)_k e^{ikt}$$

converges for all $t \in \mathbb{R}$, and it follows from the assumption $(a_j)^*_{-k} = (a_j)_k$ that $\gamma_j(t) \in \mathbb{R}$. Furthermore, it follows from Lemma 10.5.11 that $\hat{\gamma}'_j(t) = \sum_{k \in \mathbb{Z}} ik(a_j)_k e^{ikt}$ for all $t \in \mathbb{R}$. From (10.59) and the relation (**??**) between the nonlinearities f and ϕ, we infer that $f_j(\hat{\gamma}(t)) = \sum_{k \in \mathbb{Z}} (\phi_j(a))_k e^{ikt}$ for all $t \in \mathbb{R}$. It now follows from (10.19) that $\hat{\gamma}'(t) = \tau_0 f(\hat{\gamma}(t))$ for $t \in \mathbb{R}$. A time rescaling shows that γ given by (10.18) is a periodic orbit of the vector field f with period $2\pi\tau_0$. $\square$

10.2.1 Projections and truncations

The main differences between the current setting of Fourier series for periodic solutions and Taylor series for initial value problems in Chapter 8 are the two-sidedness of the sequences and the extra unknown period as well as the phase conditions equation. This leads to small adaptations in the notation, but overall the similarities with Chapter 8 are striking.

As the Banach space which contains the unknowns we work with

$$X = X^\nu \stackrel{\text{def}}{=} \mathbb{C} \times \left(\ell^1_\nu\right)^n \tag{10.20}$$

which we endow with the norm

$$\|x\|_X = \|(\tau_0, a)\| \stackrel{\text{def}}{=} \max\{|\tau_0|, \|a_1\|_{1,\nu}, \|a_2\|_{1,\nu}, \ldots, \|a_n\|_{1,\nu}\}. \tag{10.21}$$

For $N \in \mathbb{N}$ we define the projection

$$(\pi_N^{1,\nu} a)_k \stackrel{\text{def}}{=} \begin{cases} a_k & \text{for } |k| \leq N, \\ 0 & \text{for } |k| > N, \end{cases} \qquad \text{for any } a \in \ell^1_\nu.$$

We extend this projection to X by setting

$$\pi_N x \stackrel{\text{def}}{=} (\tau_0, \pi_N^{1,\nu} a_1, \ldots, \pi_N^{1,\nu} a_n) \qquad \text{for any } x = (\tau_0, a) \in X,$$

and $\pi_\infty = I_X - \pi_N$ is the projection on the tail. The projections onto the components of $x = (\tau_0, a) \in X$ are denoted by

$$\pi^0 x = \tau_0 \qquad \text{and} \qquad \pi^j x = a_j \quad \text{for } j = 1, \ldots, n.$$

We also use $\pi^j a = a_j$ for $j = 1, \ldots, n$. The subspace $X_N = \pi_N X$ can be identified with $\mathbb{C} \times (\mathbb{C}^{2N+1})^n \simeq \mathbb{C}^{\tilde{N}}$, with $N_* = n(2N+1) + 1$ The natural bijection is denoted by $\iota_N : X_N \to \mathbb{C}^{N_*}$. In this context we denote elements of $\mathbb{C}^{N_*}$ by

$$x^{(N)} = (\tau_0, a^{(N)}), \qquad \text{with } a^{(N)} = (a_1^{(N)}, \ldots, a_n^{(N)}) \quad \text{and } a_j^{(N)} = ((a_j)_{-N}, \ldots, (a_j)_N).$$

The complement of X_N is denoted by $X_\infty = \pi_\infty X$. Finally, on $\pi_\infty \ell^1_\nu$ we introduce

$$(\Lambda_N a)_k = \begin{cases} 0 & \text{if } |k| \leq N, \\ \frac{1}{ik} a_k & \text{if } |k| \geq N+1, \end{cases} \tag{10.22}$$

which has norm $\|\Lambda_N\|_{B(\pi_\infty \ell^1_\nu)} = \frac{1}{N+1}$.

10.2.2 Fixed point formulation

We define the finite dimensional truncation $F^{(N)} : \mathbb{C}^{N_*} \to \mathbb{C}^{N_*}$ of the full map F by

$$F^{(N)}(x^{(N)}) = \iota_N \pi_N F(\iota_N^{-1} x^{(N)}) \quad \text{for } x^{(N)} \in \mathbb{C}^{N_*}.$$

In practice we find $\bar{x}$ by soving numerically for an approximate zero $\bar{x}^{(N)}$ of $F^{(N)}$. We then denote by $\bar{x} = \iota_N^{-1} \bar{x}^{(N)}$ the approximate zero of F.

Using the identification of X_N and $\mathbb{C}^{N_*}$ via ι_N, as well as the (finite dimensional) Jacobian $DF^{(N)}$, we define the operator $A_N^\dagger : X_N \to X_N$ by

$$A_N^\dagger \stackrel{\text{def}}{=} \iota_N^{-1} DF^{(N)}(\bar{x}^{(N)}) \iota_N,$$

and it follows that

$$A_N^\dagger = \pi_N DF(\bar{x}) \quad \text{for any } \bar{a} \in X_N. \tag{10.23}$$

We also define the operator $A_\infty^\dagger : X_\infty \to X_\infty$ by

$$(\pi^j A_\infty^\dagger y)_k = ik(\pi^j y)_k, \qquad \text{for } j = 1, \ldots, n. \tag{10.24}$$

We now set

$$A^\dagger y = A_N^\dagger \pi_N y + A_\infty^\dagger \pi_\infty y. \tag{10.25}$$

With $A^{(N)}$ a numerically computed approximate inverse of the matrix $DF^{(N)}(\bar{x}^{(N)})$, we set $A_N : X_N \to X_N$ to be the approximate inverse of $A_N^\dagger$ defined by

$$A_N = \iota_N^{-1} A^{(N)} \iota_N.$$

We define $A_\infty : X_\infty \to X_\infty$ by

$$(\pi^j A_\infty y)_k = \frac{1}{ik}(\pi^j y)_k, \qquad \text{for } j = 1, \ldots, n.$$

so that $A_\infty = (A_\infty^\dagger)^{-1}$, hence it makes sense to write A_∞^{-1} instead of $A_\infty^\dagger$. We then define

$$Ay = A_N \pi_N y + A_\infty \pi_\infty y. \tag{10.26}$$

Decomposition into components

A natural decomposition of $A^\dagger$ is given by

$$\pi^j A^\dagger y = A_{j,j'}^\dagger \pi^{j'} y,$$

or in matrix form:

$$A^\dagger \stackrel{\text{def}}{=} \begin{bmatrix} A_{0,0}^\dagger & A_{0,1}^\dagger & \cdots & A_{0,n}^\dagger \\ A_{1,0}^\dagger & A_{1,1}^\dagger & \cdots & A_{1,n}^\dagger \\ \vdots & \vdots & \ddots & \vdots \\ A_{n,0}^\dagger & A_{n,1}^\dagger & \cdots & A_{n,n}^\dagger \end{bmatrix}. \tag{10.27}$$

The component linear operators $A_{j,j'}^\dagger$ map between different spaces depending on whether $j, j' = 0$ or $j, j' = 1, \ldots, n$.

For $1 \leq j, j' \leq N$ we may interpret $A_{j,j'}^\dagger$ as a bi-infinite matrix

$$A_{j,j'}^\dagger \stackrel{\text{def}}{=} \begin{bmatrix} \ddots & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -i(N+2)\delta_{j,j'} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -i(N+1)\delta_{j,j'} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & D_{a_{j'}} F_j^{(N)}(\bar{x}^{(N)}) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & i(N+1)\delta_{j,j'} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & i(N+2)\delta_{j,j'} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \ddots \end{bmatrix},$$

and similarly for $A_{j,j'}$, but with the reciprocal on the diagonal tails.

The Jacobian

In order to perform the necessarry estimates, it is useful to have explicit expressions for the derivative of the components of F. Analogous to Section 8.2.1, we write

$$D\phi_j(a)b = \sum_{j'=1}^{n} D_{a_{j'}}\phi_j(a)b_{j'} = \sum_{j'=1}^{n}\sum_{|\alpha|=1}^{M}(c_\alpha)_j\alpha_{j'}a^{\alpha-\epsilon_{j'}} * b_{j'}.$$

Denoting $D_{j'} = D_{a_{j'}}$, for $1 \le j, j' \le n$ the derivative $D_{j'}F_j(x)$ can be characterized by its action on natural basis vectors $\{e_m\}_{m\in\mathbb{Z}}$ of ℓ^1_ν, defined by $(e_m)_k = \delta_{m,k}$. Writing $x = (\tau_0, a)$ throughout we obtain

$$(D_{j'}F_j(x)e_m)_k = ik\delta_{j',j}\delta_{m,k} - \tilde{\tau}(D_{j'}\phi_j(a))_{k-m}. \tag{10.28}$$

Concerning the derivative of the phase condition (10.15), we obtain

$$D_{j'}F_0(x)e_m = \begin{cases} \hat{\xi} & \text{for } |m| \le N_0 \\ 0 & \text{for } |m| > N_0. \end{cases}$$

For simplicity we will assume from now on that $N_0 = N$. Finally, denoting $D_0F = D_{\tau_0}F$, the remaining partial derivatives are

$$\begin{aligned} &D_0F_0(x) = 0 \\ &D_0F_j(x) = -\phi_j(a) \qquad \text{for } j = 1, \ldots, n. \end{aligned}$$

Estimates

With A defined in (10.26), we are ready to define the Newton-like fixed point operator

$$T(x) = x - AF(x).$$

We leave it to the reader to check that $AF(x) \in X$ for all $x \in X$, cf. Proposition 8.2.6.

We now compute the bounds Y_0, Z_0, Z_1 and Z_2 satisfying (7.30), (7.31), (7.32) and (7.33), respectively.

<u>The bound Y_0</u>

Lemma 10.2.4. *Set*

$$\begin{aligned} Y_0^{(0)} &\stackrel{\text{def}}{=} \left| A_{0,0}^{(N)}F_0(\bar{x}) + \sum_{n=1}^{N} A_{0,i}^{(N)} \cdot F_n^{(N)}(\bar{x}) \right| \\ Y_0^{(j)} &\stackrel{\text{def}}{=} \sum_{|k|<K} \left| \sum_{j'=0}^{N} \left(A_{j,j'}^{(N)}F_{j'}^{(N)}(\bar{x}) \right)_k \right| \nu^{|k|} + \sum_{|k|=N+1}^{MN} \left| \frac{\bar{\tau}_0}{k}(\phi_j(\bar{a}))_k \right| \nu^{|k|}, \end{aligned}$$

for $j = 1, \dots n$, *and*

$$Y_0 \stackrel{\text{def}}{=} \max\left(Y_0^{(0)}, Y_0^{(1)}, \dots, Y_0^{(N)}\right). \tag{10.29}$$

Then $\|[T(\bar{x}) - \bar{x}]\|_X \le Y_0$.

Proof. For the first (or zeroth) component we obtain

$$|[T(\bar{x}) - \bar{x}]_0| = |[AF(\bar{x})]_0| \le \left|\sum_{j'=0}^{n} A_{0,j'} F_{j'}(\bar{x})\right| = Y_0^{(0)},$$

where the computation of the expression $Y_0^{(0)}$ consists only of evaluating finite sums.

Next, since $f\colon \mathbb{R}^n \to \mathbb{R}^n$ is a polynomial vector field of degree M and $\bar{a} \in \pi^N(\ell^1_\nu)^n$, we infer that $(\phi(\bar{a}))_k$ vanishes for $|k| > MN$. Since $(F_j(\bar{x}))_k = -\bar{\tau}_0(\phi_j(\bar{a}))_k$ for $|k| > N$ for $j = 1, \dots, n$, we obtain

$$\|[T(\bar{x}) - \bar{x}]_j\|_{1,\nu} = \|[-AF(\bar{x})]_j\|_{1,\nu} = \left\|\sum_{j'=0}^{n} A_{j,j'} F_{j'}(\bar{x})\right\|_{1,\nu} = Y_0^{(j)},$$

where the expression $Y_0^{(j)}$ is again a collection of finite sums that can be evaluated with interval arithmetic. Finally, it follows from the definition of the norm on X that $\|T(\bar{x}) - \bar{x}\|_X$ is bounded by Y_0. □

The bound Z_0

The analysis of $I - AA^\dagger$ is the same as in Section 8.2.1. The only difference with Lemma 8.2.8 is that we need to treat $j = 0$ differently from $j = 1, \dots, n$. We introduce $B = I - AA^\dagger$ and $B^{(N)} = I_{\mathbb{C}^{N_*}} - A^{(N)} DF^{(N)}(\bar{x}^{(N)})$. We use a decomposition analogous to the one in (10.27).

Lemma 10.2.5. *Set*

$$Z_0^{(0)} \stackrel{\text{def}}{=} |B_{0,0}^{(N)}| + \sum_{j'=1}^{n} \max_{|k| \le N} \frac{1}{\nu^{|k|}} |(B_{0,j'}^{(N)})_k|,$$

$$Z_0^{(j)} \stackrel{\text{def}}{=} \sum_{|k| \le N} |(B_{j,0}^{(N)})_k| \nu^{|k|} + \sum_{j'=1}^{n} \max_{|k'| \le N} \frac{1}{\nu^{|k'|}} \sum_{|k| \le N} |(B_{j,j'}^{(N)})_{k,k'}| \nu^{|k|},$$

for $j = 1, \dots, n$, *and*

$$Z_0 \stackrel{\text{def}}{=} \max\left(Z_0^{(0)}, Z_0^{(1)}, \dots, Z_0^{(n)}\right). \tag{10.30}$$

Then $\|I - AA^\dagger\|_{B(X)} \le Z_0$.

Proof. The proof is analogous to that of Lemma 8.2.8. The only changes are the expressions for the norms of $B^{(N)}_{0,0}$, $B^{(N)}_{0,j'}$ and $B^{(N)}_{j,0}$ for $1 \le j, j' \le N$. □

The bound Z_1

Given any $h \in X$ with $\|h\|_X \le 1$ we set

$$z_{m,j'} = (D_{j'} F_m(\bar{x}) - A^\dagger)\pi^{j'} h.$$

By using that $\pi_N DF(\bar{a})\pi_N = A^\dagger_N \pi_N$, see (10.23) and (10.25), we then decompose

$$\begin{aligned} DF(\bar{a}) - A^\dagger &= (\pi_N + \pi_\infty) DF(\bar{a}) - (A^\dagger_N \pi_N + A^\dagger_\infty \pi_\infty) \\ &= \pi_N DF(\bar{a})(\pi_N + \pi_\infty) - A^\dagger_N \pi_N + \pi_\infty DF(\bar{a}) - A^\dagger_\infty \pi_\infty \\ &= \pi_N DF(\bar{a})\pi_\infty + \pi_\infty (DF(\bar{a}) - A^\dagger_\infty \pi_\infty). \end{aligned}$$

The expression for the Jacobian then implies that

$$(\pi_N z_{m,j'})_k = \begin{cases} -(\bar{\tau}_0 \pi_N D_{j'} \phi_m(\bar{a}) \pi_\infty \pi^{j'} h)_k & \text{for } 1 \le m, j' \le n, \\ 0 & \text{for } m = 0 \text{ or } j = 0, \end{cases}$$

and

$$(\pi_\infty z_{m,j'})_k = \begin{cases} -(\bar{\tau}_0 \pi_\infty D_{j'} \phi_m(\bar{a}) \pi^{j'} h)_k & \text{for } 1 \le m, j' \le n, \\ -(\pi_\infty \phi_m(\bar{a}))_k \pi^0 h & \text{for } j' = 0 \text{ and } 1 \le m \le n, \\ 0 & \text{for } m = 0. \end{cases} \tag{10.31}$$

To simplify the notation in the next lemma, we generalize the notation from (**??**) to cover also nonsquare matrices $M \in M_{2N_1+1, 2N_2+1}(\mathbb{C})$:

$$\|M\|_{1,\nu} = \max_{|j| \le N_2} \frac{1}{\nu^{|j|}} \sum_{|k| \le N_1} |M_{k,j}| \nu^{|k|}. \tag{10.32}$$

We note that for $N_2 = 0$ this is the regular norm on ℓ^1_ν, for $N_1 = 0$ it is the dual norm (the norm on $\ell^\infty_{\nu^{-1}}$) while for $N_1 = N_2 = 0$ it is just the absolute value.

Lemma 10.2.6. *Let $\Gamma^{(N)}_\nu$ be as defined in* (10.6) *and*

$$\kappa_{j,m}(\nu) \stackrel{\text{def}}{=} \|A^{(N)}_{j,m} \Gamma^{(N)}_\nu\|_{1,\nu} \qquad \text{for } 0 \le j \le n,\ 1 \le m \le n.$$

Set

$$Z_1^{(j,j')} \stackrel{\text{def}}{=} \begin{cases} 0 & \text{for } j = j' = 0, \\ \frac{1}{N+1} \|\pi_\infty \phi_j(\bar{a})\|_{\ell^1_\nu} & \text{for } 1 \le j \le n,\ j' = 0 \\ |\bar{\tau}_0| \sum_{m=1}^n \kappa_{j,m}(\nu) \|D_{j'} \phi_m(\bar{a})\|_{\ell^1_\nu} & \text{for } 1 \le j' \le n,\ j = 0 \\ |\bar{\tau}_0| \sum_{m=1}^n \kappa_{j,m}(\nu) \|D_{j'} \phi_m(\bar{a})\|_{\ell^1_\nu} + \frac{|\bar{\tau}_0|}{N+1} \|D_{j'} \phi_j(\bar{a})\|_{\ell^1_\nu} & \text{for } 1 \le j, j' \le n, \end{cases}$$

and

$$Z_1 \stackrel{\text{def}}{=} \max_{0\le j\le n} \sum_{j'=0}^{n} Z_1^{(j,j')}. \tag{10.33}$$

Then $\|A[DF(\bar{x}) - A^\dagger]\|_{B(X)} \le Z_1$.

Proof. For $h \in X$ with $\|h\|_X \le 1$ we set

$$w_{j,j'} \stackrel{\text{def}}{=} \pi^j A[DF(\bar{x}) - A^\dagger]\pi^{j'} h = \sum_{m=0}^{n} A_{j,m} z_{m,j'}.$$

Since $\pi_N z_{m,j'}$ whenever $m = 0$ or $j' = 0$, and $\pi_\infty z_{m,j'}$ when $m = 0$, we infer that

$$\begin{aligned}
&w_{0,0} = 0 \\
&w_{j,0} = \Lambda_N \pi_\infty z_{j,0} && \text{for } 1 \le j \le n, \\
&w_{j,j'} = \sum_{m=1}^{n} (A_N)_{j,m} \pi_N z_{m,j'} + \Lambda_N \pi_\infty z_{j,j'} && \text{for } 0 \le j \le n,\ 1 \le j' \le n,
\end{aligned}$$

with Λ_N given in (10.22). By using the explicit expressions for $(\pi_\infty z_{m,j'})_k$ in (10.31) we estimate, for $1 \le j \le n$

$$\|\Lambda_N \pi_\infty z_{j,j'}\|_{\ell^1_\nu} \le \|\Lambda\|_{B(\pi_\infty \ell^1_\nu)} \|z_{j,j'}\|_{\ell^1_\nu} \le \begin{cases} \frac{1}{N+1} \|\pi_\infty \phi_j(\bar{a})\|_{\ell^1_\nu} & \text{for } j' = 0, \\ \frac{|\bar{\tau}_0|}{N+1} \|D_{j'} \phi_j(\bar{a})\|_{\ell^1_\nu} & \text{for } 1 \le j' \le n. \end{cases}$$

For the remaining terms we use the notation for the norm from (10.32) and apply Lemma **??** to estimate, for $1 \le m, j' \le n$ and $0 \le j \le n$,

$$\begin{aligned}
\|(A_N)_{j,m} \pi_N z_{m,j'}\|_{1,\nu} &\le \|(A_N)_{j,m} \Gamma_\nu^{(N)} (\Gamma_\nu^{(N)})^{-1} \pi_N z_{m,j'}\|_{1,\nu} \\
&\le \|(A_N)_{j,m} \Gamma_\nu^{(N)}\|_{1,\nu} \|(\Gamma_\nu^{(N)})^{-1} z_{m,j'}\|_{\ell^1_\nu} \\
&= \kappa_{j,m}(\nu) \|(\Gamma_\nu^{(N)})^{-1} \bar{\tau}_0 D_{j'} \phi_m(\bar{a}) \pi_\infty \pi^{j'} h\|_{\ell^1_\nu} \\
&\le \kappa_{j,m}(\nu) |\bar{\tau}_0| \|D_{j'} \phi_m(\bar{a})\|_{\ell^1_\nu}.
\end{aligned}$$

Putting the estimates together finishes the proof. $\square$

The bound Z_2

For the Z_2 bound we proceed as for the Z_2 bound in Section 8.2.1, essentially educing it to an estimate of the second derative of T in a ball of radius r_* around $\bar{x}$. Given arbitrary $h \in X$ with $\|h\|_X \le 1$ and arbitrary $y \in \overline{B_r(\bar{x})}$, set

$$\beta_{j,j'} \stackrel{\text{def}}{=} [D_{j'} F_j(y) - D_{j'} F_j(\bar{x})] \pi_{j'} h \quad \text{for } j, j' = 0, \dots n.$$

By linearity of the phase condition F_0 we get that $\beta_{0,j'}$ for all $j' = 0, \ldots, n$. For $j = 1, \ldots n$ and $j' = 0, \ldots, n$ we proceed as follows to estimate $\|\beta_{j,j'}\|_{1,\nu}$. Any element y of the ball $\overline{B_r(\bar{x})}$ may be written as $y = \bar{x} + r\tilde{h}$ with $\tilde{h} \in \overline{B_1(0)}$. Applying the Mean Value Theorem yields

$$\|\beta_{j,j'}\|_{1,\nu} \leq \sum_{m=0}^{n} \sup_{\substack{a \in \overline{B_r(\bar{a})} \\ \tilde{h} \in \overline{B_1(0)}}} \left\| D_m(D_{j'} F_j(a)\pi^{j'} h)\pi^m \tilde{h} \right\|_{1,\nu} r \,. \tag{10.34}$$

The second derivatives are given by $D_0 D_0 F_j(a) = 0$ and for $1 \leq j', m \leq n$

$$\begin{aligned} D_0(D_{j'} F_j(a)\pi^{j'} h)\pi^0 \tilde{h} &= -(D_{j'}\phi_j(a)\pi^{j'} h)\pi^0 \tilde{h} \\ D_m(D_{j'} F_j(a)\pi^{j'} h)\pi^m \tilde{h} &= -\tau D^2_{j',m}\phi_j(a)\pi^{j'} h \pi^m h, \end{aligned}$$

using the product notation for the convolution. We set

$$\eta_{j,j'}(r) \stackrel{\text{def}}{=} \sum_{|\alpha|=1}^{M} |(c_\alpha)_j| \, \alpha_{j'} \prod_{l=1}^{n} (\|\bar{a}_l\|_{1,\nu} + r)^{\alpha_l - \delta_{j',l}}, \tag{10.35}$$

$$\zeta_{j,m,j'}(r) \stackrel{\text{def}}{=} \sum_{|\alpha|=2}^{M} |(c_\alpha)_j| \, \alpha_{j'}(\alpha_m - \delta_{j',m}) \prod_{l=1}^{n} (\|\bar{a}_l\|_{1,\nu} + r)^{\alpha_l - \delta_{j',l} - \delta_{m,l}}. \tag{10.36}$$

Since $h, \tilde{h} \in B_1(0)$, we obtain $\|\pi^{j'} h\|_{\ell^1_\nu} \leq 1$ and $|\pi^0 \tilde{h}| \leq 1$ and $\|\pi^m \tilde{h}\|_{\ell^1_\nu} \leq 1$. The Banach algebra structure then leads to the estimate

$$\|\beta_{j,j'}\|_{1,\nu} \leq \left[\eta_{j,j'}(r) + (|\bar{\tau}| + r) \sum_{m=1}^{n} \zeta_{j,m,j'}(r) \right] r. \tag{10.37}$$

We can now use the triangle inequality to estimate $\|A[DF(b) - DF(\bar{a})]\|_{B(X)}$. The relevant (operator) norms of the components of $A_{j,m}$ reduce to the computable expressions

$$\begin{aligned} K_{0,m} &\stackrel{\text{def}}{=} \max_{|k'| \leq N} \frac{1}{\nu^{|k'|}} |(A^{(N)}_{0,m})_{k'}| \\ K_{j,m} &\stackrel{\text{def}}{=} \max \left(\max_{|k'| \leq N} \frac{1}{\nu^{|k'|}} \sum_{|k| \leq N} |(A^{(N)}_{j,m})_{k,k'}| \nu^{|k|}, \frac{\delta_{j,m})}{N+1} \right), \end{aligned}$$

for $1 \leq j, m \leq n$.

Lemma 10.2.7. *Let $r_* > 0$. With η and ζ defined in* (10.35) *and* (10.36)*, set*

$$Z_2^{(j,j')} = \sum_{m'=1}^{n} K_{j,m'} \left[\eta_{m',j'}(r_*) + (|\bar{\tau}| + r_*) \sum_{m=1}^{n} \zeta_{m',m,j'}(r_*) \right],$$

for $0 \le j, j' \le n$, *and*

$$Z_2 \stackrel{\text{def}}{=} \max_{0 \le j \le n} \sum_{j'=0}^{n} Z_2^{(j,j')}. \tag{10.38}$$

Then $\|A[DF(y) - DF(\bar{x})]\|_{B(X)} \le Z_2 r$ *for all* $y \in \overline{B_r(\bar{x})}$ *for any* $r \in [0, r_*]$.

Proof. In terms of components we have

$$\pi^j A[DF(y) - DF(\bar{x})]h = \sum_{m'=0}^{n} A_{j,m'} \beta_{m',j'} = \sum_{m'=1}^{n} A_{j,m'} \beta_{m',j'}.$$

in view of $\beta_{0,j'} = 0$. By using the bound (10.37) on β and the bounds $\|A_{0,m'}\|_{B(\ell^1_\nu, \mathbb{C})} = K_{0,m'}$ and $\|A_{j,m'}\|_{B(\ell^1_\nu)} = K_{j,m'}$, we obtain

$$\|\pi^0 A[DF(y) - DF(\bar{x})]\|_{B(X,\mathbb{C})} \le \sum_{j'=0}^{n} Z_2^{(0,j')}$$

$$\|\pi^j A[DF(y) - DF(\bar{x})]\|_{B(X,\ell^1_\nu)} \le \sum_{j'=0}^{n} Z_2^{(j,j')}$$

for $j = 1, \ldots n$. We infer that Z_2 defined in (10.38) bounds $\|A[DF(y) - DF(\bar{x})]\|_B(X)$. $\square$

10.2.3 Verifying that we obtain real solutions

Combining the bounds Y_0, Z_0, Z_1 and Z_2 given respectively by (10.29) , (10.30) , (10.33) and (10.30), we can define the radii polynomial as

$$p(r) \stackrel{\text{def}}{=} Z_2 r^2 - (1 - Z_0 - Z_1) r + Y_0. \tag{10.39}$$

Assuming the existence of $r_0 \in (0, r_*)$ such that $p(r_0) < 0$, by Theorem 7.6.2, there exists a unique $\tilde{x} \in \overline{B_{r_0}(\bar{x})} \subset X$ such that $T(\tilde{x}) = \tilde{x}$, or equivalently such that $F(\tilde{x}) = 0$.

The true solution $\tilde{x} = (\tilde{\tau}_0, \tilde{a}_1, \ldots, \tilde{a}_n)$ is an element of the Banach space $X = \mathbb{C} \times \left(\ell^1_\nu\right)^n$, which does not directly imply that the resulting periodic solution is real. To conclude that $\tilde{x}$ does indeed correspond to a real-valued periodic orbit, we show that the solutions obtained from the radii polynomial approach are in fact in the *symmetry* space

$$X_{\text{sym}} \stackrel{\text{def}}{=} \mathbb{R} \times \left(\tilde{\ell}^1_\nu\right)^n, \tag{10.40}$$

where

$$\tilde{\ell}^1_\nu \stackrel{\text{def}}{=} \left\{ a \in \ell^1_\nu \,|\, a_{-k} = a_k^* \text{ for all } k \in \mathbb{Z} \right\}, \tag{10.41}$$

provided that the numerical approximation $\bar{x}$ satisfies $\bar{x} \in X_{\text{sym}}$. This symmetry operation interacts nicely with the convolution.

Lemma 10.2.8. *For any $a, \tilde{a} \in \ell^1_\nu$ we have $a^* * \tilde{a}^* = (a * \tilde{a})^*$.*

Proof. For any $k \in \mathbb{Z}$ we have

$$(a^* * \tilde{a}^*)_k = \sum_{m\in\mathbb{Z}} (a^*)_{k-m}(\tilde{a}^*)_m = \sum_{m\in\mathbb{Z}} a^*_{-k+m}\tilde{a}^*_{-m} = \sum_{m'\in\mathbb{Z}} (a_{-k-m'}\tilde{a}_{m'})^* = (a*\tilde{a})^*_{-k} = ((a*\tilde{a})^*)_k.$$

□

Given $x = (\omega, a_1, \ldots, a_n) \in X = \mathbb{C} \times \left(\ell^1_\nu\right)^n$, denote by $x^* = (\omega^*, a_1^*, \ldots, a_n^*) \in X$ the "complex conjugate" of x, where for $j = 1, \ldots, n$, the vector a_j^* is defined component-wise by

$$(a_j^*)_k \stackrel{\text{def}}{=} (a_j)^*_{-k}.$$

Using the above definition, the space X_{sym} defined in (10.40) may be expressed as

$$X_{\text{sym}} \stackrel{\text{def}}{=} \{x \in X \mid x^* = x\}.$$

Lemma 10.2.9. *If $x \in X$ solves $F(x) = 0$, then $F(x^*) = 0$.*

Proof. It follows from Lemma 10.2.8 and induction on the order of the polynomial that $\phi(a^*) = \phi(a)^*$, i.e., $(\phi_j(a^*))_k = (\phi_j(a))^*_{-k}$ for all $1 \le j \le n$ and $k \in \mathbb{Z}$. This implies that

$$\begin{aligned}
(F_j(x^*))_k &= ik\omega^*(a_j^*)_k - \tau_0^*(\phi_j(a^*))_k \\
&= \left(ik\omega^*(a_j)^*_{-k} - \tau_0(\phi_j(a))^*_{-k}\right) \\
&= (-ik\omega(a_j)_{-k} - \tau_0(\phi_j(a))_{-k})^* \\
&= (F_j(x))^*_{-k}.
\end{aligned}$$

For $j = 0$ we infer that, by using that $\xi_0, \hat{\xi}_0 \in \mathbb{R}^n$,

$$\begin{aligned}
F_0(x^*) &= \sum_{j=1}^n \sum_{|k|\le N_0} \hat{\xi}_{0,j}(a_j^*)_k - \sum_{j=1}^n \hat{\xi}_{0,j}\xi_{0,j} \\
&= \sum_{j=1}^n \sum_{|k|\le N_0} \hat{\xi}_{0,j}(a_j)^* {-k} - \sum_{j=1}^n \hat{\xi}_{0,j}\xi_{0,j} \\
&= \sum_{j=1}^n \sum_{|k|\le N_0} (\hat{\xi}_{0,j}(a_j)^k - \sum_{j=1}^n \hat{\xi}_{0,j}\xi_{0,j})^* \\
&= F_0(x)^*.
\end{aligned}$$

We conclude that if $F(x) = 0$ then $F(x^*) = F(x)^* = 0$. □

Lemma 10.2.10. *Let $\bar{x} \in X_{\text{sym}}$ and $r > 0$. Then $x \in \overline{B_r(\bar{x})}$ implies that $x^* \in \overline{B_r(\bar{x})}$.*

Proof. We first assert that $\|x\|_X = \|x^*\|_X$ for all $x = (\tau_0, a) \in X$. Namely $|\tau_0^*| = |\tau_0|$ and for $j = 1, \dots, n$

$$\|a_j^*\|_{\ell^1_\nu} = \sum_{k\in\mathbb{Z}} |(a_j^*)_k|\nu^{|k|} = \sum_{k\in\mathbb{Z}} |(a_j)^*_{-k}|\nu^{|k|} = \sum_{k'\in\mathbb{Z}} |(a_j)_{k'}|\nu^{|k'|} = \|a_j\|_{\ell^1_\nu}.$$

Since $\bar{x}^* = \bar{x}$, we then infer that $\|x^* - \bar{x}\|_X = \|x^* - \bar{x}^*\|_X = \|(x-\bar{x})^*\| - X = \|x - \bar{x}\|_X$. □

Lemma 10.2.11. *Let $\bar{x} X_{\mathrm{sym}}$. Assume $Z_2 r^2 - (1 - Z_0 - Z_1)r + Y_0 < 0$ for some $r \in [0, r_*]$, and let $\tilde{x}$ be the unique fixed point of T in $\overline{B_r(\bar{x})}$. Then $\tilde{x} \in X_{\mathrm{sym}}$.*

Proof. This follows from Lemmas 10.2.9 and 10.2.10 and uniqueness of the fixed point of T in $\overline{B_r(\bar{x})}$. □

Conclusion

We have thus proven the following result.

Theorem 10.2.12. *Let $N \in \mathbb{N}$ and $\nu > 0$. Let f be a polynomial given by* (10.11). *Let $\bar{x} \in X_N \cap X_{\mathrm{sym}}$. Let Y_0, Z_0, Z_1 and $Z_2(r_*)$ be defined by* (10.29), (10.30), (10.33) *and* (10.38), *respectively. Assume that $r_0 \in (0, r_*)$ satisfies the inequality*

$$Z_2(r_*)r_0^2 + (Z_0 + Z_1 - 1)r_0 + Y_0 < 0.$$

Then there is an $\tilde{x} = (\tilde{\tau}_0, \tilde{a}) \in X_{\mathrm{sym}}$ with $\|\tilde{x} - \bar{x}\|_X \le r_0$ such that

$$\tilde{\gamma}(t) = \sum_{k\in\mathbb{Z}} \tilde{a}_k e^{ikt/\tilde{\tau}_0} \tag{10.42}$$

is a periodic orbit of the vector field f with period $2\pi\tilde{\tau}_0$.

We note that injectivity of A follows from the natural analogue of Lemma 8.2.5. Concerning the error bound between the numerical apprroximation

$$\bar{\gamma}(t) = \sum_{|k|\le\mathbb{N}} \bar{a}_k e^{ikt/\bar{\tau}_0}$$

and the periodic solution $\tilde{\gamma}(t)$ given in (10.42), we have for the period that $|2\pi\tilde{\tau}_0 - 2\pi\bar{\tau}_0| \le 2\pi r_0$, and for distance between the orbits in phase space we have

$$\|\tilde{\gamma}(t) - \bar{\gamma}(\tfrac{\bar{\tau}_0}{\tilde{\tau}_0}t)\| \le r_0 \qquad \text{for all } t \in \mathbb{R}.$$

10.3 Periodic solutions in the Lorenz system

We recall that the Lorenz system is given by

$$\begin{cases} \dot{x}_1 = \sigma(x_2 - x_1) \\ \dot{x}_2 = \rho x_1 - x_2 - x_1 x_3 \\ \dot{x}_3 = -\beta x_3 + x_1 x_2, \end{cases} \tag{10.43}$$

where we again consider with classical parameter values $\sigma = 10$, $\rho = 28$ and $\beta = 8/3$. We applied the general method from Section 10.2 to the Lorenz system to obtain validated periodic orbits, see Figure 10.3. We used the computational parameters $N = 600$, $\nu = 1.05$ and $r_* = 10^{-5}$.

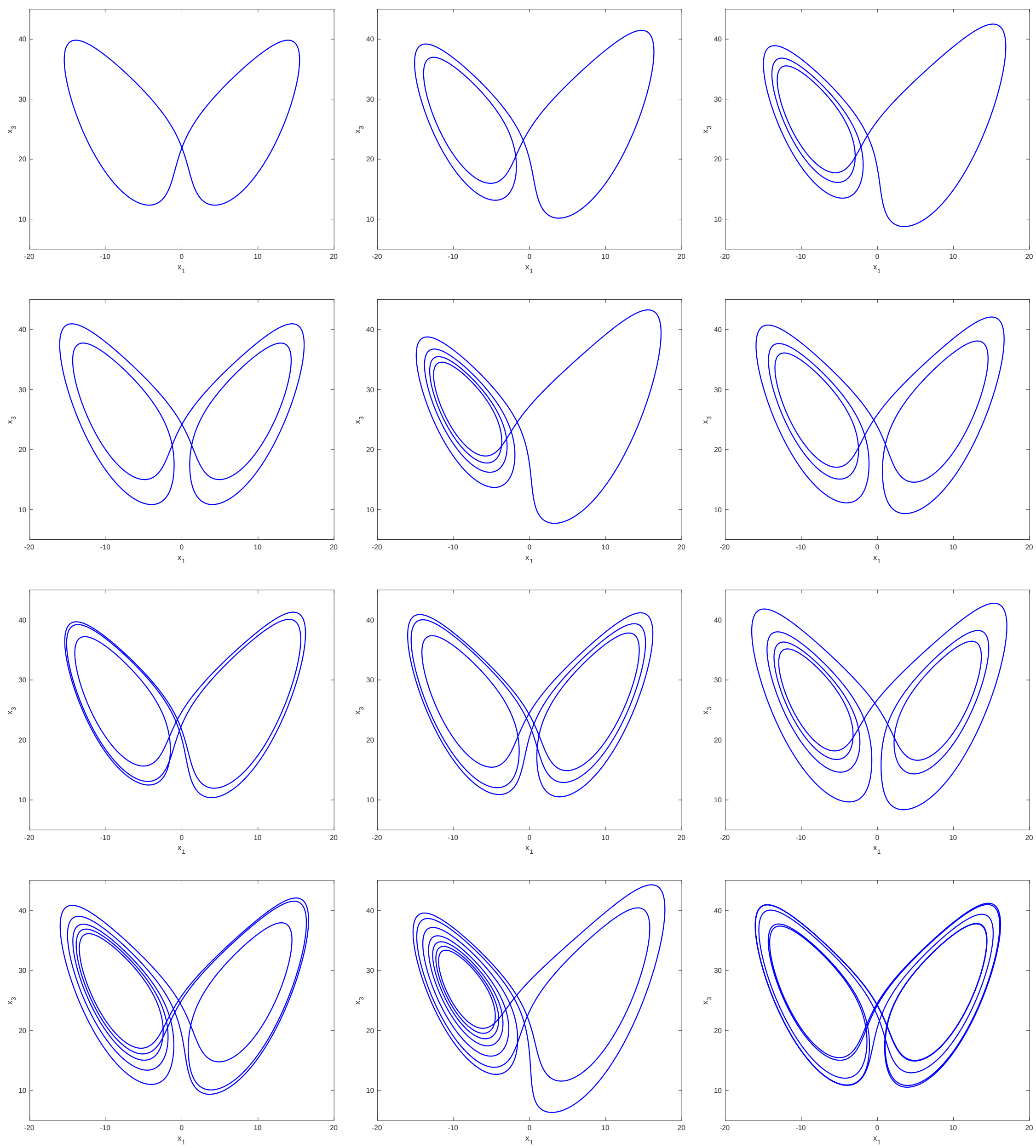

Figure 10.3: Twelve validated periodic orbits in the Lorenz system. The blue curves truncated Fourier series with $N = 600$ terms. For some orbits the C^0 difference between the true and approximate solution is about $r = 7 \times 10^{-9}$ while for others it is about $r = 1.2 \times 10^{-7}$. The periods of these orbits varies from about 1.5578 to about 7.7212.

10.4 Hopf Bifurcation

In this section, we introduce the concept of a Hopf bifurcation, which is a phenomenon whereby a family of periodic orbits is born from a branch of equilibria.

Theorem 10.4.1 (Hopf Bifurcation Theorem). *Let $f\colon \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ and consider the parameter dependent vector field*

$$\dot{x} = f(x, \lambda). \tag{10.44}$$

We assume that (10.44) *possesses an analytic family $x = x(\lambda)$ of equilibrium solutions, that is $f(x(\lambda), \lambda) = 0$ for all $\lambda \in (\lambda^-, \lambda^+)$. We suppose that for a certain value $\tilde{\lambda} \in (\lambda^-, \lambda^+)$, the Jacobian matrix $D_x f(x(\tilde{\lambda}), \tilde{\lambda})$ has two purely imaginary eigenvalues $\pm i\tilde{\omega}$ (with $\tilde{\omega} \neq 0$) and no other eigenvalue of $D_x f(x(\tilde{\lambda}), \tilde{\lambda})$ is an integral multiple of $i\tilde{\omega}$. If $\alpha(\lambda) + i\omega(\lambda)$ is the continuation of the eigenvalue $i\tilde{\omega}$ (that is with $\alpha(\tilde{\lambda}) = 0$ and $\omega(\tilde{\lambda}) = \tilde{\omega}$), then we assume that*

$$\alpha'(\tilde{\lambda}) \neq 0. \tag{10.45}$$

Then there exist continuous functions $\lambda = \lambda(\epsilon)$ and $\tau = \tau(\epsilon)$ depending on a parameter ϵ with $\lambda(0) = \tilde{\lambda}$, $\tau(0) = \frac{2\pi}{\omega}$ such that there are nonconstant periodic solutions $x(t, \epsilon)$ of (10.44) *with period $\tau(\epsilon)$ which collapse into $x(\tilde{\lambda})$ as $\epsilon \to 0$.*

Proof. See Section 3C of [25]. □

Definition 10.4.2. When all hypotheses of Theorem 10.4.1 are satisfied, we say that there is a *Hopf bifurcation* at the point $(x, \lambda) = (\tilde{x}(\lambda^*), \lambda^*)$.

As in the case of saddle-node bifurcations (see Theorem 4.6.7), we introduce a zero-finding problem, whose non-degenerate zeros (plus an eigenvalue condition) correspond to a Hopf bifurcation.

At a Hopf bifurcation point, we get to solve the problem

$$\begin{aligned} f(x, \lambda) &= 0 \\ g(x, v, \lambda, \omega) &\stackrel{\text{def}}{=} Df(x, \lambda)v - i\omega v = 0. \end{aligned}$$

We turn to the real valued formulation of the problem. Hence, let $v_1 = \mathrm{Re}(v)$ and $v_2 = \mathrm{Im}(v)$ be the real and the imaginary parts of v, respectively. Similarly, set

$$\begin{aligned} \mathrm{Re}(g(x, v_1 + iv_2, \lambda, \omega)) &= \mathrm{Re}(Df(x, \lambda)(v_1 + iv_2) - i\omega(v_1 + iv_2)) = Df(x, \lambda)v_1 + \omega v_2 \\ \mathrm{Im}(g(x, v_1 + iv_2, \lambda, \omega)) &= \mathrm{Im}(Df(x, \lambda)(v_1 + iv_2) - i\omega(v_1 + iv_2)) = Df(x, \lambda)v_2 - \omega v_1. \end{aligned}$$

Letting $X = (x, v_1, v_2, \lambda, \omega) \in \mathbb{R}^{3n+2}$, define the *Hopf map*

$$F(X) \stackrel{\text{def}}{=} \begin{pmatrix} f(x, \lambda) \\ Df(x, \lambda)v_1 + \omega v_2 \\ Df(x, \lambda)v_2 - \omega v_1 \\ v_1 \cdot \xi_1 - 1 \\ v_2 \cdot \xi_2 - 1 \end{pmatrix}, \tag{10.46}$$

where the conditions $v_i \cdot \xi_i = 1$ are phase conditions for the vectors $v_1, v_2 \in \mathbb{R}^n$ (with $\xi_i \neq 0$). Typically, $\xi_i = \bar{v}_i \neq 0$ $(i = 1, 2)$ is an approximation for v_i. This choice of phase condition is linear and mimics the more standard nonlinear phase condition $\|v_i\|^2 - 1 = 0$.

Theorem 10.4.3. *Assume that $f\colon \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ is analytic, that $\tilde{X} = (\tilde{x}, \tilde{v}_1, \tilde{v}_2, \tilde{\lambda}, \tilde{\omega})$ is a nondegenerate zero of the Hopf map F defined in* (10.46)*, and that $\tilde{\omega} \neq 0$. Then $\pm i\tilde{\omega}$ are purely imaginary eigenvalues of $Df(\tilde{x}, \tilde{\lambda})$. Assume that the other $n-2$ eigenvalues of $D_x f(\tilde{x}, \tilde{\lambda})$ have nonzero real parts. Then $(\tilde{x}, \tilde{\lambda})$ is a Hopf bifurcation point.*

Proof. Let $\tilde{X} = (\tilde{x}, \tilde{v}_1, \tilde{v}_2, \tilde{\lambda}, \tilde{\omega}) \in \mathbb{R}^{3n+2}$ be such that $F(\tilde{X}) = 0$. Recalling the definition of the Hopf map in (10.46), one gets that

$$D_X F(\tilde{X}) = \begin{pmatrix} D_x f(\tilde{x}, \tilde{\lambda}) & 0 & 0 & D_\lambda f(\tilde{x}, \tilde{\lambda}) & 0 \\ D_x^2 f(\tilde{x}, \tilde{\lambda})\tilde{v}_1 & D_x f(\tilde{x}, \tilde{\lambda}) & \tilde{\omega} I & D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v}_1 & \tilde{v}_2 \\ D_x^2 f(\tilde{x}, \tilde{\lambda})\tilde{v}_2 & -\tilde{\omega} I & D_x f(\tilde{x}, \tilde{\lambda}) & D_{x,\lambda} f(\tilde{x}, \tilde{\lambda})\tilde{v}_2 & -\tilde{v}_1 \\ 0 & \xi_1^T & 0 & 0 & 0 \\ 0 & 0 & \xi_2^T & 0 & 0 \end{pmatrix}. \tag{10.47}$$

Note that $\tilde{v}_1, \tilde{v}_2 \neq 0$ since $\tilde{v}_1 \cdot \xi_1 = \tilde{v}_2 \cdot \xi_2 = 1 \neq 0$. Moreover, by hypothesis, all eigenvalues of $D_x f(\tilde{x}, \tilde{\lambda})$ are nonzero. This implies that $D_x f(\tilde{x}, \tilde{\lambda})$ is invertible. By analyticity of f and by the Implicit Function Theorem, there exists an analytic family $x = x(\lambda)$ of equilibrium solutions, that is $f(x(\lambda), \lambda) = 0$ for all $\lambda \in (\lambda^-, \lambda^+)$ with $\tilde{x} = x(\tilde{\lambda})$. Differentiating $f(x(\lambda), \lambda) = 0$ leads to

$$D_x f(\tilde{x}, \tilde{\lambda})\dot{x} + D_\lambda(\tilde{x}, \tilde{\lambda}) = 0, \qquad \text{where } \dot{x} \stackrel{\text{def}}{=} \frac{dx}{d\lambda}(\tilde{\lambda}). \tag{10.48}$$

For all $\lambda \in (\lambda^-, \lambda^+)$, denote by $\alpha(\lambda) + i\omega(\lambda) \in \sigma(D_x f(\tilde{x}(\lambda), \lambda) = 0$ the continuation of the eigenvalue $i\tilde{\omega} \in \sigma(D_x f(\tilde{x}, \tilde{\lambda}))$. Hence, $\alpha(\tilde{\lambda}) = 0$ and $\omega(\tilde{\lambda}) = \tilde{\omega}$. Similarly, denote by $v_1(\lambda) + iv_2(\lambda)$ the eigenvector associated to $\alpha(\lambda) + i\omega(\lambda)$ which is the continuation of the eigenvector $\tilde{v}_1 + i\tilde{v}_2$. Hence, $v_1(\tilde{\lambda}) = \tilde{v}_1$ and $v_2(\tilde{\lambda}) = \tilde{v}_2$.

Without loss of generality, we may assume that for all $\lambda \in (\lambda^-, \lambda^+)$,

$$v_1(\lambda) \cdot \xi_1 = v_2(\lambda) \cdot \xi_2 = 1,$$

which implies that $v_1'(\lambda) \cdot \xi_1 = v_2'(\lambda) \cdot \xi_2 = 0$. In particular, at $\lambda = \tilde{\lambda}$,

$$\dot{v}_1 \cdot \xi_1 = \dot{v}_2 \cdot \xi_2 = 0, \qquad \text{where } \dot{v}_i \stackrel{\text{def}}{=} \frac{dv_i}{d\lambda}(\tilde{\lambda}). \tag{10.49}$$

Now, for all $\lambda \in (\lambda^-, \lambda^+)$

$$Df(x(\lambda), \lambda)(v_1(\lambda) + iv_2(\lambda)) = (\alpha(\lambda) + i\omega(\lambda))(v_1(\lambda) + iv_2(\lambda)),$$

which leads to

$$\begin{aligned} D_x f(x(\lambda),\lambda)v_1(\lambda) &= \alpha(\lambda)v_1(\lambda) - v_2(\lambda)\omega(\lambda) \\ D_x f(x(\lambda),\lambda)v_2(\lambda) &= \alpha(\lambda)v_2(\lambda) + v_1(\lambda)\omega(\lambda). \end{aligned}$$

Differentiating the last two equalities with respect to λ, and evaluating at $\lambda = \tilde{\lambda}$ yields

$$\begin{aligned} D_x^2 f(\tilde{x},\tilde{\lambda})(\dot{x},\tilde{v}_1) + D_{x,\lambda} f(\tilde{x},\tilde{\lambda})\tilde{v}_1 + D_x f(\tilde{x},\tilde{\lambda})\dot{v}_1 &= \alpha'(\tilde{\lambda})\tilde{v}_1 + \alpha(\tilde{\lambda})\dot{v}_1 - \dot{v}_2\omega(\tilde{\lambda}) - \tilde{v}_2\omega'(\tilde{\lambda}) \\ D_x^2 f(\tilde{x},\tilde{\lambda})(\dot{x},\tilde{v}_2) + D_{x,\lambda} f(\tilde{x},\tilde{\lambda})\tilde{v}_2 + D_x f(\tilde{x},\tilde{\lambda})\dot{v}_2 &= \alpha'(\tilde{\lambda})\tilde{v}_2 + \alpha(\tilde{\lambda})\dot{v}_2 + \dot{v}_1\omega(\tilde{\lambda}) + \tilde{v}_1\omega'(\tilde{\lambda}). \end{aligned}$$

Since $\alpha(\tilde{\lambda}) = 0$ and $\omega(\tilde{\lambda}) = \tilde{\omega}$, then

$$\alpha'(\tilde{\lambda})\tilde{v}_1 = D_x^2 f(\tilde{x},\tilde{\lambda})(\dot{x},\tilde{v}_1) + D_{x,\lambda} f(\tilde{x},\tilde{\lambda})\tilde{v}_1 + D_x f(\tilde{x},\tilde{\lambda})\dot{v}_1 + \dot{v}_2\tilde{\omega} + \tilde{v}_2\omega'(\tilde{\lambda}) \tag{10.50}$$

$$\alpha'(\tilde{\lambda})\tilde{v}_2 = D_x^2 f(\tilde{x},\tilde{\lambda})(\dot{x},\tilde{v}_2) + D_{x,\lambda} f(\tilde{x},\tilde{\lambda})\tilde{v}_2 + D_x f(\tilde{x},\tilde{\lambda})\dot{v}_2 - \dot{v}_1\tilde{\omega} - \tilde{v}_1\omega'(\tilde{\lambda}). \tag{10.51}$$

Assume that the *crossing condition* (10.45) does not hold, that is $\alpha'(\tilde{\lambda}) = 0$, and denote

$$X^* \stackrel{\text{def}}{=} \begin{pmatrix} \dot{x} \\ \dot{v}_1 \\ \dot{v}_2 \\ 1 \\ \omega'(\tilde{\lambda}) \end{pmatrix} \neq 0.$$

Then, combining (10.47), (10.48), (10.49), (10.50), (10.51), and the assumption that $\alpha'(\tilde{\lambda}) = 0$,

$$D_X F(\tilde{X})X^* = \begin{pmatrix} D_x f(\tilde{x},\tilde{\lambda})\dot{x} + D_\lambda f(\tilde{x},\tilde{\lambda}) \\ D_x^2 f(\tilde{x},\tilde{\lambda})(\dot{x},\tilde{v}_1) + D_x f(\tilde{x},\tilde{\lambda})\dot{v}_1 + \tilde{\omega}\dot{v}_2 + D_{x,\lambda} f(\tilde{x},\tilde{\lambda})\tilde{v}_1 + \omega'(\tilde{\lambda})\tilde{v}_2 \\ D_x^2 f(\tilde{x},\tilde{\lambda})(\dot{x},\tilde{v}_2) - \tilde{\omega}\dot{v}_1 + D_x f(\tilde{x},\tilde{\lambda})\dot{v}_2 + D_{x,\lambda} f(\tilde{x},\tilde{\lambda})\tilde{v}_2 - \tilde{v}_1\omega'(\tilde{\lambda}) \\ \dot{v}_1 \cdot \xi_1 \\ \dot{v}_2 \cdot \xi_2 \end{pmatrix} = 0$$

which violates the nondegeneracy of $\tilde{X}$. Hence, the condition (10.45) holds.

Finally, by hypothesis, no other eigenvalue (besides of $\pm i\tilde{\omega}$) $D_x f(\tilde{x}(\lambda^*),\lambda^*)$ is an integral multiple of $i\tilde{\omega}$.

By Theorem 10.4.1, we conclude that $(\tilde{x},\tilde{\lambda})$ is a Hopf bifurcation point. $\square$

10.4.1 Hopf bifurcation in the Lorenz equations

Consider the Lorenz equations (10.43) with $\sigma = 10$, $\beta = 8/3$ and leave ρ as a parameter. In this case, $n = 3$ and therefore the *Hopf map* F defined in (10.46) satisfies $F : \mathbb{R}^{11} \to \mathbb{R}^{11}$.

We applied Newton's method and obtained the point

$$\bar{X} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \bar{x}_3 \\ (\bar{v}_1)_1 \\ (\bar{v}_1)_2 \\ (\bar{v}_1)_3 \\ (\bar{v}_2)_1 \\ (\bar{v}_2)_2 \\ (\bar{v}_2)_3 \\ \bar{\rho} \\ \bar{\omega} \end{pmatrix} = \begin{pmatrix} 7.956019457871823 \\ 7.956019457871823 \\ 23.736842105263147 \\ 0.556151966545753 \\ 0.696054940405531 \\ 0.454095265384198 \\ 0.145360836252368 \\ -0.389909295949055 \\ 0.909307961153164 \\ 24.736842105263147 \\ -9.624530063715754 \end{pmatrix} \tag{10.52}$$

such that $F(\bar{X}) \approx 0$. Using the Localized radii polynomials in finite dimensions (version 1), as presented in Theorem 2.5.1, we proved the existence of a unique nondegenerate zero $\tilde{X} \in B_{1.1\times 10^{-13}}(\bar{X})$ of the Hopf map. Then, we set $\tilde{x} = (\tilde{X}_1, \tilde{X}_2, \tilde{X}_3) = (\tilde{x}_1, \tilde{x}_2, \tilde{x}_3)$ and $\tilde{\rho} = \tilde{X}_{10}$. Using the control that

$$|\tilde{\rho} - \bar{\rho}|, \|\tilde{x} - \bar{x}\|_\infty \leq 1.1 \times 10^{-13},$$

we verified that the other eigenvalue $\tilde{\lambda}_1$ of $Df(\tilde{x})$ satisfies

$$\tilde{\lambda}_1 \in [-13.66666666666682, -13.66666666666651]$$

and therefore the other eigenvalue is not on the imaginary axis. By Theorem 10.4.3, $(\tilde{x}, \tilde{\lambda})$ is a Hopf bifurcation point.

10.5 Fourier series and the Fourier transform

Any differentiable 2π-periodic function g can be written in terms of a Fourier series (see [9] for a proof).

Proposition 10.5.1. *Let $g\colon [0, 2\pi] \to \mathbb{C}$ be a differentiable periodic function, i.e. $g(0) = g(2\pi)$. Then*

$$g(t) = \sum_{k\in\mathbb{Z}} a_k e^{ikt}, \qquad \text{for all } t \in [0, 2\pi],$$

where

$$a_k = \frac{1}{2\pi} \int_0^{2\pi} g(s) e^{-iks}\, ds, \qquad k \in \mathbb{Z}.$$

In light of the earlier chapters this suggests working with the Banach space of *two*-sided sequences (see Example 7.3.4)

$$\ell^1_{\nu,\mathbb{Z}} \stackrel{\text{def}}{=} \{a = \{a_k\}_{k\in\mathbb{Z}} : a_k \in \mathbb{C} \text{ and } \|a\|_{1,\nu} < \infty\}, \tag{10.53}$$

where

$$\|a\|_{1,\nu} \stackrel{\text{def}}{=} \sum_{k\in\mathbb{Z}} |a_k|\nu^{|k|}. \tag{10.54}$$

Throughout this chapter we will write ℓ^1_ν for $\ell^1_{\nu,\mathbb{Z}}$.

To be able to work in these spaces with geometrically decaying Fourier coefficients, we need to show that periodic solutions to analytic ODEs can be represented by sequences in ℓ^1_ν. With this in mind we make a brief detour into the realm of complex analysis.

Recall the *identity principle* (also called the *uniqueness principle*) for analytic functions: that if two analytic functions $g, \tilde{g} \colon U \to \mathbb{C}$ have $g(z) = \tilde{g}(z)$ for all $z \in S$ – where $S \subset U$ is any set having an accumulation point – then $g = \tilde{g}$ on U. This follows from the fact that zeros of analytic functions are isolated.

Definition 10.5.2. A function $g \colon (a, b) \to \mathbb{R}$ is *real analytic* if for every $t \in (a, b)$ there is an $R > 0$, called the *radius of convergence* so that the power series

$$g(t) = \sum_{k=0}^{\infty} \frac{g^{(k)}(t_0)}{k!}(t - t_0)^k,$$

converges absolutely and uniformly for all $|t - t_0| < R$.

Note that, if g is real analytic, then the power series

$$G(z) = \sum_{k=0}^{\infty} \frac{g^{(k)}(t_0)}{k!}(z - t_0)^k,$$

defines a complex analytic function on the disk $D_R(t_0) \subset \mathbb{C}$ having $G(t) = g(t)$ for all $t \in (t_0 - R, t_0 + R)$. Moreover, note that if $\tilde{G} \colon D_R(t_0) \to \mathbb{C}$ is any other analytic function with $\tilde{G}(t) = g(t)$ for $t \in (t_0 - R, t_0 + R)$, then $G(t) = \tilde{G}(t)$ on $(t_0 - R, t_0 + R)$, a set which accumulates in $D_R(t_0)$. Then, by the identity principle, $G(z) = \tilde{G}(z)$ on $D_R(t_0)$. This uniqueness justifies our calling G *the* complex analytic extension of g at t_0. Letting

$$\Omega = \bigcup_{t_0\in(a,b)} D_{R_{t_0}}(t_0),$$

we see that a real analytic function can always be extended uniquely to a complex analytic function on some open set $\Omega \subset \mathbb{C}$ having $(a, b) \subset \Omega$.

Given $\rho > 0$, consider the open complex strip

$$A_\rho \stackrel{\text{def}}{=} \{z \in \mathbb{C} : |\operatorname{Im}(z)| < \rho\}.$$

A function $g \colon A_\rho \to \mathbb{C}$ is 2π-*periodic* on A_ρ if $g(z + 2\pi) = g(z)$ for all $z \in A_\rho$.

Lemma 10.5.3. *Let $g\colon \mathbb{R} \to \mathbb{R}$ be a real analytic periodic function of period 2π. Then there exists $\rho > 0$ such that g can be extended to an analytic function defined on the complex strip A_ρ. Moreover, the analytic extension is 2π-periodic on A_ρ.*

Proof. Consider $t_1 \in \mathbb{R}$ and let $t_2 = t_1 + 2\pi$. Since g is real analytic, the power series expansions at t_1 and also at t_2 converge absolutely and uniformly to g and have some non-zero radii of convergence, say R_1 and R_2 respectively. Let $R_0 = \min(R_1, R_2)$ and define the functions

$$G_1(z) = g(z),$$

and

$$G_2(z) = g(z + 2\pi),$$

for $z \in D_{R_0}(t_1)$ (here we identify g with its complex analytic extension by power series). Note that, since g is periodic on the real line, we have

$$G_1(s) = g(s) = g(s + 2\pi) = G_2(s),$$

for all $s \in (t_1 - R_0, t_1 + R_0) = I$. But the set I has an accumulation point in $D_{R_0}(t_1)$ (indeed, each $s \in I$ is such a point) so it follows from the identity principle that $G_1 = G_2$ on $D_{R_0}(t_1)$. In particular these functions have the same power series expansions, with the same radius of convergence. Then in fact, $R_1 = R_2$. Moreover, if $z \in D_{R_1}(t_1)$, then $z + 2\pi \in D_{R_1}(t_2)$, and since $G_1 = G_2$ on $D_{R_1}(t_1)$ we have that

$$g(z + 2\pi) = G_2(z) = G_1(z) = g(z),$$

for all $z \in D_{R_1}(t_1)$. Since t_1 was arbitrary, we have that the analytic extension of g is 2π-periodic. Finally, consider $t \in [0, 2\pi]$ and let R_t denote the radius of the analytic extension at t. Then

$$[0, 2\pi] \subset \bigcup_{t \in [0,2\pi]} D_{R_t}(t).$$

By compactness there is a finite collection $t_1, \ldots, t_N \in [0, 2\pi]$ so that

$$[0, 2\pi] \subset \bigcup_{k=1}^{N} D_{R_k}(t_k),$$

where $R_k \stackrel{\text{def}}{=} R_{t_k}$. Since this finite collection of disks covers $[0, 2\pi]$ there is a $\rho > 0$ so that

$$z \in \bigcup_{k=1}^{N} D_{R_k}(t_k),$$

whenever $\mathrm{real}(z) \in [0, 2\pi]$ and $|\mathrm{imag}(z)| < \rho$. Moreover, by the periodicity we have that

$$[2\pi n, 2\pi(n+1)] \subset \bigcup_{k=1}^{N} D_{R_k}(t_k + 2\pi n),$$

for all $n \in \mathbb{Z}$. Then

$$A_\rho \subset \bigcup_{n\in\mathbb{Z}} \bigcup_{k=1}^{N} D_{R_k}(t_k + 2\pi n),$$

so that g is analytic in A_ρ. □

We know from Proposition 19.6.7 that when the vector field is analytic, any solution is real analytic. For the special case of a 2π-periodic solution of (10.9), Lemma implies that it can be extended analytically to A_ρ for some $\rho > 0$. The following theorem implies that the Fourier coefficients of such a periodic solution have geometric decay.

Theorem 10.5.4. [Paley-Wiener Theorem] *Suppose that $g\colon \mathbb{R} \to \mathbb{R}$ is a real analytic 2π-periodic function. Consider the Fourier expansion of u*

$$g(t) = \sum_{k\in\mathbb{Z}} a_k e^{ikt}.$$

Let $a \stackrel{\text{def}}{=} \{a_k\}_{k\in\mathbb{Z}}$. Then, there exists $\nu > 1$ such that $a \in \ell^1_\nu$.

Proof. By Lemma 10.5, there exists a $\rho > 0$ such that u can be extended to an analytic function on the strip A_ρ of width ρ. Choose any $\epsilon \in (0, \rho)$. The fact that u is analytic on A_ρ implies that g is continuous, and hence bounded on the closed strip $\overline{A_{\rho-\epsilon}}$. Then there is a $C = C(g, \epsilon) > 0$ such that

$$\sup_{z\in\overline{A_{\rho-\epsilon}}} |g(z)| \le C.$$

To prove the theorem we need to obtain bounds on the k-th Fourier coefficient

$$a_k = \frac{1}{2\pi}\int_0^{2\pi} g(s)e^{-iks}\, ds$$

of g. To do this consider the function

$$u_k(z) \stackrel{\text{def}}{=} \frac{1}{2\pi} g(z)e^{-ikz}.$$

Note that $u_k(z)$ is analytic and 2π-periodic on A_ρ as e^{-ikz} is entire and 2π-periodic. Consider $k \ge 0$. Let $\Gamma = \Gamma_1 + \Gamma_2 + \Gamma_3 + \Gamma_4$ be the closed contour shown in Figure 10.4.

By Cauchy's integral theorem, the line integral of an analytic function along a closed curve is zero and hence

$$\int_\Gamma u_k(z)\, dz = 0.$$

Since $u_k(z)$ is 2π-periodic on A_ρ

$$\int_{\Gamma_2} u_k(z)\, dz = -\int_{\Gamma_4} u_k(z)\, dz.$$

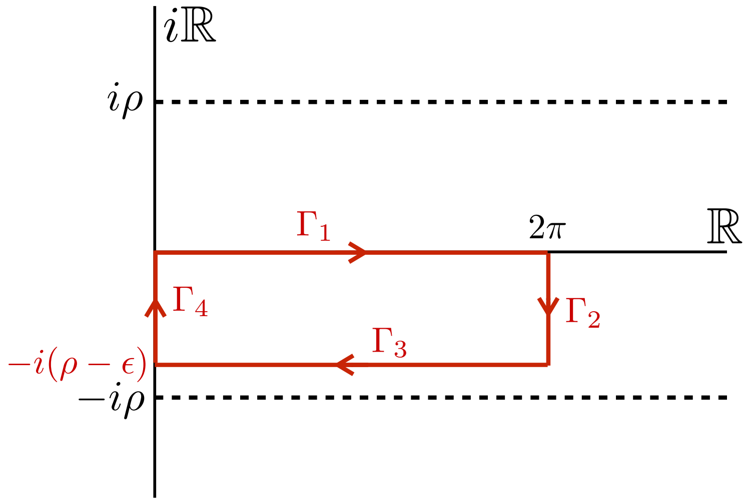

Figure 10.4: The contour Γ for the line integral used to bound of $|a_k|$ when $k \geq 0$. For the case $k < 0$, consider the contour obtained by reflecting Γ about the real axis.

Thus,

$$\int_{\Gamma_1} u_k(z)\,dz = -\int_{\Gamma_3} u_k(z)\,dz.$$

Parameterizing $\Gamma_1\colon [0, 2\pi] \to \mathbb{C}$ and $\Gamma_3\colon [0, 2\pi] \to \mathbb{C}$ by

$$\Gamma_1(s) = s, \quad \text{and} \quad \Gamma_3(s) = 2\pi - i(\rho - \epsilon) - s,$$

we have on the one hand that

$$\int_{\Gamma_1} u_k(z)\,dz = \int_0^{2\pi} u_k(\Gamma_1(s))\Gamma_1'(s)\,ds = \frac{1}{2\pi}\int_0^{2\pi} g(s)e^{-iks}\,ds = a_k,$$

while on the other hand

$$-\int_{\Gamma_3} u_k(z)\,dz = -\int_0^{2\pi} u_k(\Gamma_3(s))\Gamma_3'(s)\,ds = \frac{1}{2\pi}\int_0^{2\pi} g(2\pi - s - i(\rho-\epsilon))e^{-ik(2\pi-s)}e^{-k(\rho-\epsilon)}\,ds.$$

Thus, for $k \geq 0$,

$$\begin{aligned} |a_k| &= \left|\int_{\Gamma_1} u_k(z)\,dz\right| \\ &= \left|\int_{\Gamma_3} u_k(z)\,dz\right| \\ &\leq \frac{1}{2\pi}\int_0^{2\pi} \left|g(2\pi - s - i(r-\epsilon))e^{-ik(2\pi-s)}\right| e^{-k(\rho-\epsilon)}\,ds \\ &\leq Ce^{-k(\rho-\epsilon)}. \end{aligned} \tag{10.55}$$

Similarly for $k < 0$ consider the contour obtained by "flipping" γ about the real axis, i.e. take $\gamma = \Gamma_1 + \Gamma_2 + \Gamma_3 + \Gamma_4$ with Γ_1 as before but

$$\Gamma_3(s) = 2\pi - s + i(\rho - \epsilon),$$

noting that the integrals over Γ_2 and Γ_4 cancel out one another as before. Repeating the computations for this new contour gives that

$$|a_k| \leq Ce^{k(\rho-\epsilon)}, \qquad \text{for all } k < 0. \tag{10.56}$$

Combining Equations (10.55) and (10.56) gives that

$$|a_k| \leq Ce^{-|k|(\rho-\epsilon)}, \quad \text{for all } k \in \mathbb{Z}.$$

Let $\tilde{\nu} \stackrel{\text{def}}{=} e^{(\rho-\epsilon)} > 1$. Then, for all $k \in \mathbb{Z}$,

$$|a_k| \leq \frac{C}{\tilde{\nu}^{|k|}}.$$

Choose any $\nu > 1$ such that

$$\nu < \tilde{\nu} = e^{(\rho-\epsilon)} \tag{10.57}$$

Then

$$\|a\|_{1,\nu} = \sum_{k\in\mathbb{Z}} |a_k|\nu^{|k|} \leq C \sum_{k\in\mathbb{Z}} \left(\frac{\nu}{\tilde{\nu}}\right)^{|k|} < \infty$$

since $0 < \frac{\nu}{\tilde{\nu}} < 1$. This implies that $a = \{a_k\}_{k\in\mathbb{Z}} \in \ell^1_\nu$. □

Remark 10.5.5. It follows from the proof of Theorem 10.5.4, and in particular (10.57), that if g is analytic on A_ρ, then its Fourier coefficients $a = \{a_k\}_{k\in\mathbb{Z}}$ lie in ℓ^1_ν for any $\nu < e^\rho$.

Theorem 10.5.4 justifies looking for periodic solutions in a Banach space X which consists of a product of the space of two-sided sequences ℓ^1_ν given in (10.54).

We are also interested in a converse of the Paley-Wiener Theorem (Theorem 10.5.4). The following theorem says that if the Fourier coefficients of a function decay exponentially fast then the function is analytic on some strip. Note that if $a = \{a_k\}_{k\in\mathbb{Z}} \in \ell^1_\nu$ with $\nu > 1$ then we have such decay, as

$$|a_k|\nu^{|k|} \leq \|a\|_{1,\nu} = \sum_{k\in\mathbb{Z}} |a_k|\nu^{|k|} < \infty,$$

for each $k \in \mathbb{Z}$, which is to say that

$$|a_k| \leq \frac{\|a\|_{1,\nu}}{\nu^{|k|}} = Ce^{-|k|r}, \qquad \text{with } C \stackrel{\text{def}}{=} \|a\|_{1,\nu} \text{ and } r \stackrel{\text{def}}{=} \ln\nu > 0. \tag{10.58}$$

Theorem 10.5.6. *If a sequence of complex numbers $\{a_k\}_{k\in\mathbb{Z}}$ satisfies $|a_k| \leq Ce^{-|k|r}$, then $g(z) = \sum_{k\in\mathbb{Z}} a_k e^{ikz}$ is an analytic function on the complex strip A_r.*

The proof that we present is based on Morera's Theorem as the concepts illustrated here are useful to us again later. Recall that Morera's theorem provides a converse to the Cauchy theorem and says that if $\Omega \subset \mathbb{C}$ is an open set and if $G\colon \Omega \to \mathbb{C}$ is a continuous complex valued function such that

$$\int_\Gamma g(z)\,dz = 0,$$

for every piecewise C^1 curve homologous to zero in Ω, then g is analytic in Ω. With this result in hand we can now prove Theorem 10.5.6.

Proof. Since $|a_k| \le Ce^{-|k|r}$, the series

$$g(z) = \sum_{k\in\mathbb{Z}} a_k e^{ikz}, \quad z \in A_r.$$

converges for all $z \in A_r$. For any $\epsilon \in (0, r)$ convergence of this Fourier series is uniform on $A_{r-\epsilon}$, hence g is continuous on A_r.

Now let $\Gamma\colon [0,1] \to \mathbb{C}$ be a piecewise C^1 curve in A_r. The image of Γ is a compact set and is hence bounded away from ∂A_r. Again there is an $\epsilon > 0$ so that

$$|\mathrm{imag}(\Gamma(s))| \le r - \epsilon, \quad \forall s \in [0,1]$$

and from this we obtain (as before) that

$$\left| a_k e^{ik\Gamma(s)} \right| \le C \left(\frac{1}{e^\epsilon}\right)^{|k|}, \quad \forall s \in [0,1].$$

Then

$$|g(\Gamma(s))| = \left|\sum_{k\in\mathbb{Z}} a_k e^{ik\Gamma(s)}\right| \le C \sum_{k\in\mathbb{Z}} \left(\frac{1}{e^\epsilon}\right)^{|k|} \le \frac{2Ce^\epsilon}{e^\epsilon - 1} < \infty,$$

which is a uniform bound in s. Then the line integral

$$\int_\Gamma g(z)\,dz = \int_\Gamma \sum_{k\in\mathbb{Z}} a_k e^{ikz}\,dz = \sum_{k\in\mathbb{Z}} a_k \int_\Gamma e^{ikz}\,dz = \sum_{k\in\mathbb{Z}} a_k 0 = 0,$$

where we exchange the infinite sum with the integral due to the uniform convergence of the series and where each of the integrals

$$\int_\Gamma e^{ikz}\,dz = 0,$$

by Cauchy's Theorem, i.e. because e^{ikz} is analytic on A_r (in fact entire). The claim that g is analytic on A_r now follows from Morera's Theorem. $\square$

Analogous to the Taylor transform $\mathcal{T}$ in Section 8.4 we now introduce the Fourier transform $\mathcal{F}$. Let

$$C^\omega(A_\rho) = \{g\colon A_\rho \to \mathbb{C} \mid g \text{ is bounded and analytic}\}.$$

Theorem 10.5.4 and Remark 10.5.5 imply that the following definition makes sense.

Definition 10.5.7. The *Fourier transform* is the linear operator $\mathcal{F}_{\rho,\nu}\colon C^\omega(A_\rho) \to \ell^1_\nu$, $1 < \nu < e^\rho$ that maps $g \in C^\omega(A_\rho)$ to the Fourier coefficients of g.

Conversely, Theorem 10.5.6 and Equation (10.58) imply that the inverse transformation is well-defined in the following sense.

Definition 10.5.8. The *inverse Fourier transform* is the linear operator $\mathcal{F}^{-1}_{\rho,\nu}\colon \ell^1_\nu \to C^\omega(A_\rho)$, $0 < \rho < \log\nu$ given by

$$\mathcal{F}^{-1}_{\rho,\nu}(a) = \sum_{k\in\mathbb{Z}} a_k e^{ikz}.$$

As in the case of the Taylor transform, $\mathcal{F}^{-1}_{\rho,\nu}$ is not the inverse of $\mathcal{F}_{\rho,\nu}$, since the domains and codomains do not match. Nevertheless, at the level of manipulating convergent Fourier series, $\mathcal{F}$ and $\mathcal{F}^{-1}$ do act as inverse operations, and we will drop the subscripts from the notation.

The discrete convolution on bi-infinite sequences, as defined in (7.20), satisfies the Banach algebra property on ℓ^1_ν. In full analogy with (8.49) the relation between this convolution and the Fourier transform can be expressed as

$$\mathcal{F}(\mathcal{F}^{-1}(a)\cdot\mathcal{F}^{-1}(b)) = a * b, \tag{10.59}$$

for any $a, b \in \ell^1_\nu$ with $\nu > 1$.

Lemma 10.5.9. *Given $\nu \geq 1$, $k \in \mathbb{Z}$ and $a \in \ell^1_{\nu,\mathbb{Z}}$, the function $l^k_a\colon \ell^1_{\nu,\mathbb{Z}} \to \mathbb{C}$ defined by*

$$l^k_a(h) \stackrel{\text{def}}{=} (a * h)_k$$

with $h \in \ell^1_{\nu,\mathbb{Z}}$, is a bounded linear functional, and

$$\|l^k_a\|_{B(\ell^1_{\nu,\mathbb{Z}},\mathbb{C})} = \sup_{\|h\|_{1,\nu}\leq 1} \left|l^k_a(h)\right| \leq \sup_{j\in\mathbb{Z}} \frac{|a_{k-j}|}{\nu^{|j|}} < \infty. \tag{10.60}$$

Fix a truncation mode to be K. Given $h \in \ell^1_{\nu,\mathbb{Z}}$, set

$$\begin{aligned} h^{(K)} &\stackrel{\text{def}}{=} (\ldots, 0, 0, h_{-K+1}, \ldots, h_{K-1}, 0, 0, \ldots) \in \ell^1_{\nu,\mathbb{Z}} \\ h^{(I)} &\stackrel{\text{def}}{=} h - h^{(K)} \in \ell^1_{\nu,\mathbb{Z}}. \end{aligned}$$

Corollary 10.5.10. *Let $N \in \mathbb{N}$ with $N \geq K-1$, and let $\bar{\alpha} = (\ldots, 0, 0, \bar{\alpha}_{-N}, \ldots, \bar{\alpha}_N, 0, 0, \ldots) \in \ell^1_{\nu,\mathbb{Z}}$. Suppose that $|k| < K$ and define $\hat{l}^k_{\bar{\alpha}} \in (\ell^1_{\nu,\mathbb{Z}})^*$ by*

$$\hat{l}^k_{\bar{\alpha}}(h) \stackrel{\text{def}}{=} (\bar{\alpha} * h^{(I)})_k.$$

Then, for all $h \in \ell^1_{\nu,\mathbb{Z}}$ such that $\|h\|_{1,\nu} \leq 1$,

$$\left|\hat{l}^k_{\bar{\alpha}}(h)\right| \leq \Psi_k(\bar{\alpha}) \stackrel{\text{def}}{=} \max\left(\max_{k-N \leq j \leq -K} \frac{|\bar{\alpha}_{k-j}|}{\nu^{|j|}}, \max_{K \leq j \leq k+N} \frac{|\bar{\alpha}_{k-j}|}{\nu^{|j|}}\right). \tag{10.61}$$

Since we deal with differential equations, we will need the following result on derivatives.

Lemma 10.5.11 (Derivative of Fourier Series). *Let $a \in \ell^1_\nu$ for some $\nu > 1$. Let $g = \mathcal{F}^{-1}(a)$. Then $(\mathcal{F}(g'))_k = ika_k$ and $\mathcal{F}(g') \in \ell^1_{\nu'}$ for any $1 < \nu' < \nu$.*

Proof. We leave the proof as an exercise. □

10.6 Exercises